








Modelling Financial Markets

Chapter 1: Stylised Facts and Returns



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Unpredictable Returns, Predictable Volatility

What you should be able to do

- compute and aggregate simple and log returns
- measure tails with moments, quantiles and the Hill estimator
- tell zero correlation apart from independence
- choose tests whose reference distribution still holds
- estimate volatility from the open, high, low and close
- check a hypothesis proposed by an artificial intelligence (AI) assistant

Definition

- A **stylised fact** is a qualitative statistical regularity of **returns**
  - seen across assets, markets and periods Cont (2001)
- Each stylised fact remains a hypothesis, re-tested on each market and frequency Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
- Key message: returns show little linear autocorrelation
  - but their **size** is highly predictable

- Stylised facts are the benchmark every model of returns must reproduce (proofs: ▸ Appendix)
  - the i.i.d. (independent and identically distributed) Normal-distribution model fails most of them

The Stylised Facts of Cont (2001)

Group	Facts
Distribution	2. heavy tails; 3. gain/loss asymmetry; 4. aggregational Gaussianity; 7. conditional heavy tails
Dependence of returns	1. absence of linear autocorrelation
Dependence of volatility	5. intermittency; 6. volatility clustering; 8. slow decay of the autocorrelation of $\lvert r_t \rvert$; 11. asymmetry in time scales
Links to other variables	9. leverage effect; 10. volume/volatility correlation

- We re-test them on five markets, with data up to September 2026
  - we add the inference each test needs
- Early evidence: Mandelbrot (1963), Fama (1965); textbook treatment: Taylor (2008), Franke, Härdle & Hafner (2019)

Five Markets with Different Quoting Rules and Calendars

Series	Price used	Type, currency	Calendar	Sample	T
S&P 500	daily close (plus open, high, low)	price index, USD	US trading days	1990–2026	9 245
BET-TR	daily close	total return, RON	trading days of the Bucharest Stock Exchange (BVB)	Sep 2014 – 2026	2 992
Bitcoin	daily close	price, USD	every day, 24/7	Sep 2014 – 2026	4 384
EUR/RON	reference rate of the National Bank of Romania (BNR)	RON per EUR, managed float	Romanian working days	Jul 2005 – 2026	5 352
Gold	daily close	spot, USD per ounce	weekdays only	1990–2026	9 559

- Daily closes up to 18 September 2026; BET-TR: the BVB's BET index with dividends reinvested (total return)
  - daily log returns on each series' own calendar
  - T: the number of daily returns in the series

When Do Quotes Distort Measured Returns?

- **Managed float**: a market-determined exchange rate with central-bank interventions
  - the official BNR fixing replaces the EUR/RON market (interbank) quotes
  - in several years these had a lag-1 autocorrelation of -0.4 to -0.7, which came from quote errors

Simple or Log Return: What Adds Up over Time and What Adds Up across Assets

	Simple return R_t	**Log return r_t**
Definition	$(P_t + D_t)/P_{t-\mathbf{1}} - \mathbf{1}$	$\ln(1 + R_t) = \ln P_t - \ln P_{t-\mathbf{1}}$ (no dividend)
Over k periods	$\prod_{j=\mathbf{0}}^{k-\mathbf{1}}(1 + R_{t-j}) - \mathbf{1}$	$\sum_{j=\mathbf{0}}^{k-\mathbf{1}} r_{t-j}$: additive over time
Across assets	$R_{p,t} = \sum_i w_i R_{i,t}$: exact	only approximately
Range	$R_t \geq -100\%$	$(-\infty, +\infty)$
Use	portfolios, performance reporting	single-asset dynamics, central limit theorem (CLT)

- The simple return is the relative change in the value of the investment; the log return is the change in the log price
- Notation
  - P_t: the closing price on day t; D_t: the dividend paid in $(t-1, t]$
  - k: the number of periods aggregated; the index $j = 0, \ldots, k-1$ runs over them
  - $R_{i,t}$: the simple return of asset i; w_i: its weight in portfolio value, with $\sum_i w_i = 1$; $R_{p,t}$: the portfolio return
- With dividends use total-return or adjusted series (BET-TR)

Why the Log Return Is Always Below the Simple Return

- The log return is always below the simple return: $r = \ln(1 + R) < R$ for every $R \neq 0$
  - the reason: $\ln(1 + x) < x$ for every $x > -1$, $x \neq 0$, because the logarithm is concave
- The gap is negligible for small moves and large for big falls:
  - $R = +1\%$ gives $r = 0.995\%$
  - $R = -50\%$ gives $r = -69.3\%$
- Aggregation over time: the k-day log return is the sum of the k daily log returns
  - the k-day simple return needs the product of the growth factors $1 + R_{t-j}$
- Example: the price goes from 100 to 110, then to 99 (Seminar 1, A1)
  - the simple returns $+10\%$ and -10% sum to 0, yet the value falls by 1%: $1.1 \times 0.9 = 0.99$
  - the log returns 9.53% and -10.54% sum to $-1.01\% = \ln 0.99$
- Textbooks: Campbell, Lo & MacKinlay (1997), ch. 1; Franke, Härdle & Hafner (2019)

From Day to Year: When the $\sqrt{A}$ Rule Holds

- Annualisation turns the daily mean and volatility into values per year
- Notation
  - A: the number of return intervals per year; 252 for stocks and EUR/RON (trading days), 365 for crypto (traded every day), ≈ 260 for gold (the actual weekday quotes)
  - $\bar{r} = \frac{1}{T}\sum_{t=1}^{T} r_t$: the mean daily log return
  - $\hat{\sigma}$: the sample standard deviation of daily returns
- $\mu_{\text{ann}} = A\,\bar{r}$: the annual mean of **log** returns (the log growth rate); needs only a constant mean
  - the Sharpe ratio later uses the annual mean of **simple** returns, $A\bar{R}$, which is larger
- $\sigma_{\text{ann}} = \sqrt{A}\,\hat{\sigma}$: the annual volatility; exact only for serially uncorrelated returns with constant variance
  - then daily variances add up, and the annual variance is $A\,\hat{\sigma}^2$
  - with autocorrelation the $\sqrt{A}$ factor is no longer correct
  - EUR/RON, lag-1 autocorrelation 0.17: annual volatility rises from 4.4% to 5.3% (Appendix; Seminar 1, A7)

CAGR and Volatility Drag: Why the Arithmetic Mean Overstates Growth

- **CAGR** (Compound Annual Growth Rate): the constant annual return that takes the price from P_0 to P_T

$$\text{CAGR} = (P_T/P_0)^{1/Y} - 1$$

  - P_0, P_T: the price at the start and at the end of the sample; Y: the number of years
  - CAGR is the annual geometric mean of the growth factors

- On the log scale, the compound growth rate is the arithmetic mean minus half the variance:

$$\ln(1 + \text{CAGR}) = \tfrac{1}{Y} \ln(P_T/P_0) \approx \mu_{\text{arith}} - \sigma^2/2$$

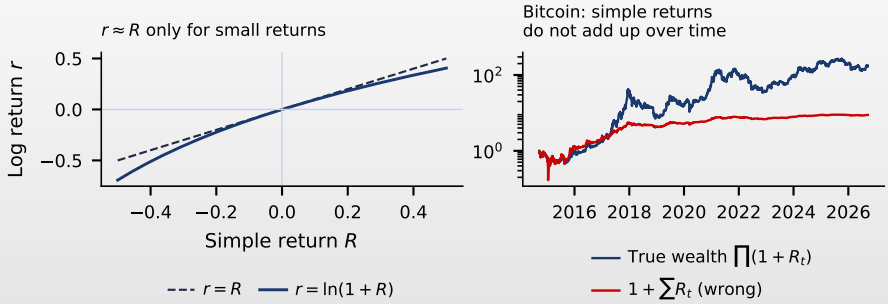
  - μ_{arith}: the annual arithmetic mean of simple returns; σ^2: their annual variance
  - $\ln(1 + \text{CAGR})$: the continuously compounded growth rate, i.e. the annual mean log return $A\bar{r}$
  - **volatility drag**: the gap between the arithmetic and the geometric mean; on the log scale it is $\approx \sigma^2/2$ and grows with the square of volatility

Volatility Drag in Numbers: S&P 500 and Bitcoin

	μ_{arith}	σ	$\sigma^2/2$	$\ln(1+\text{CAGR})$	CAGR
S&P 500 1990–2026 ($A = 252$)	9.95%	17.97%	1.62 pp	8.33%	8.69%
Bitcoin 2014–2026 ($A = 365$)	65.2%	66.6%	22.2 pp	43.1%	53.9%

- Log-scale drag, $\mu_{\text{arith}} - \ln(1 + \text{CAGR})$: $9.95 - 8.33 = 1.62$ pp (percentage points) and $65.2 - 43.1 = 22.1$ pp, matching $\sigma^2/2$
- Drag measured against CAGR, $\mu_{\text{arith}} - \text{CAGR}$: 1.26 pp for the S&P 500 and 11 pp for Bitcoin
  - smaller, because $\text{CAGR} = e^{\ln(1+\text{CAGR})} - 1$ lies above its log: $e^{0.431} - 1 = 53.9\%$
  - when a number is called "drag", say which of the two gaps it is
- S&P 500: $\hat{\sigma} = 1.133\%$ a day, 18.0% a year (log returns); price index, no dividends
- **Check**: gold has CAGR 6.7% and $\sigma = 16.3\%$. Estimate μ_{arith}
  - answer: $\ln 1.067 + 0.163^2/2 = 6.5\% + 1.3\% \approx 7.8\%$; compare Sharpe $0.48 \times 16.3\% \approx 7.8\%$ (appendix table)

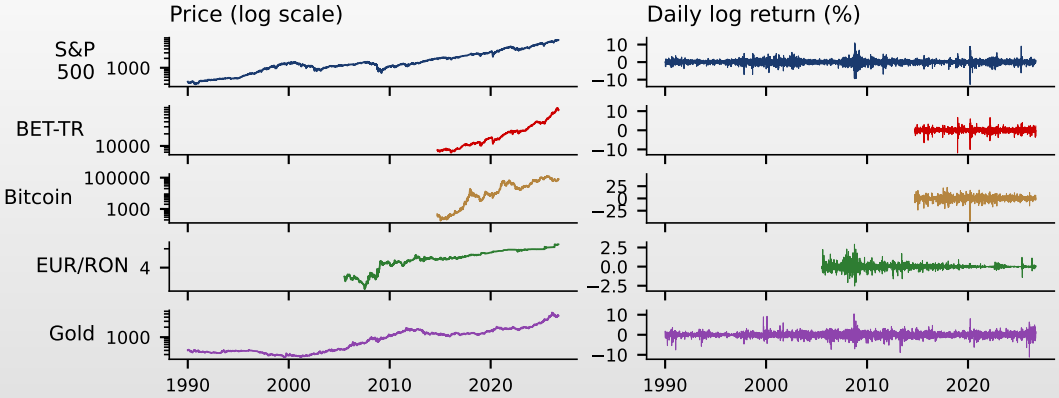
Simple and Log Returns in Practice



- Left: $r \approx R$ only for small moves
- Right: Bitcoin 2014–2026 multiplied wealth by 176.9, i.e. $\sum_t r_t = \ln 176.9 = 5.18$
  - the sum of daily log returns is $\ln(P_T/P_0)$, the log of the multiple
  - adding daily simple returns gives $1 + \sum_t R_t = 8.83$, 20 times below the true multiple

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Five Markets: Prices and Daily Log Returns



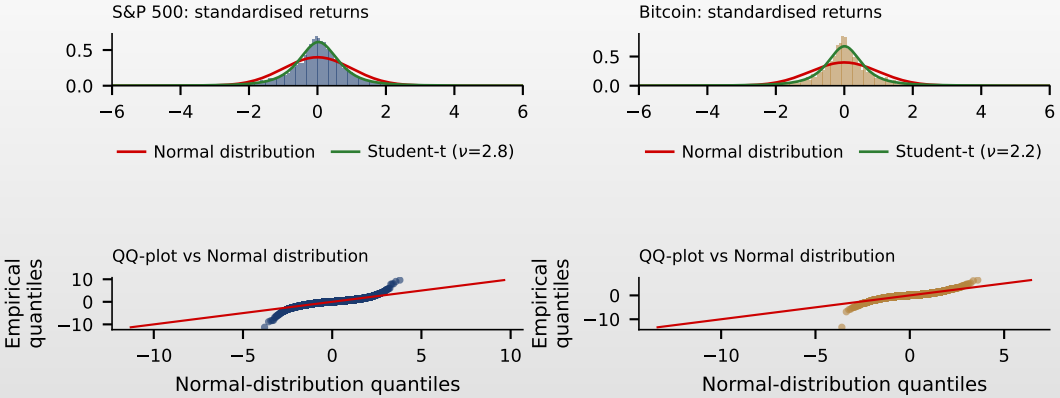
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Prices Trend, Returns Fluctuate around a Small Mean

- **Left:** prices on log scale, trending and non-stationary
- **Right:** daily log returns fluctuate around a small mean
  - calm and turbulent periods alternate
  - extremes cluster (2008, March 2020)
- EUR/RON moves more than 1% on only 1.4% of days
- Bitcoin, 12 March 2020: log return -46.5%
  - the price is multiplied by e^{r_t}, so it falls by $1 - e^{-0.465} = 37.2\%$

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Fact 2, Heavy Tails: Histograms and Quantile–Quantile (QQ) Plots



MFM_ch1_heavy_tails

Fact 2: More Extreme Days than under the Normal Distribution

- Extreme returns are far more frequent than under the Normal distribution
  - at the same variance
- Standardised returns $(r_t - \bar{r})/\hat{\sigma}$: higher peak and fatter tails
- QQ-plots compare empirical quantiles with those of the Normal distribution
  - they bend away from the line in both tails
- A Student-t fits the body of the distribution much better than the Normal distribution
  - whether it also fits the tails is a separate question, answered with the Hill estimator below

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Skewness and Kurtosis: the Shape of the Distribution after Standardisation

- Skewness and kurtosis describe the shape of the distribution, after removing its mean and scale

$$z_t = \frac{r_t - \bar{r}}{\hat{\sigma}}, \qquad \hat{S} = \frac{1}{T}\sum_{t=1}^{T} z_t^3, \qquad \hat{K} = \frac{1}{T}\sum_{t=1}^{T} z_t^4$$

  - z_t: the standardised return, with mean 0 and standard deviation 1
  - $\hat{S}$: sample skewness; $\hat{S} < 0$: the left tail is longer, large losses dominate
  - $\hat{K}$: sample kurtosis; $\hat{K} > 3$: more mass in the centre and in the tails than under the Normal distribution
- Normal distribution: $S = 0$, $K = 3$, excess kurtosis $K - 3 = 0$

How Precisely We Estimate the Mean, Skewness and Kurtosis

- **Standard error** (SE): the standard deviation of an estimator across samples
- Mean: $\mathrm{SE}(\bar{r}) = \sigma/\sqrt{T}$, with σ the standard deviation of daily returns, estimated by $\hat{\sigma}$
  - the 95% confidence interval: $\bar{r} \pm 1.96\,\mathrm{SE}(\bar{r})$
  - daily means are 13–40 times smaller than $\hat{\sigma}$
  - so, **relative to its size**, the mean is the least precisely estimated moment
  - S&P 500: $\bar{r} = 0.033\% \pm 0.023\%$ a day (95%), $T = 9{,}245$; $\mathrm{SE}/|\bar{r}| = 0.0118/0.033 \approx 0.36$
  - higher moments have a different problem: their SE may not exist at all (section on tails)
- i.i.d. data from the Normal distribution: $\sqrt{T}\,\hat{S} \to N(0,6)$ and $\sqrt{T}(\hat{K} - 3) \to N(0,24)$
  - $\to$: convergence in distribution as $T \to \infty$; $N(0,v)$: the Normal distribution with mean 0 and variance v
  - so $\mathrm{SE}(\hat{S}) \approx \sqrt{6/T}$ and $\mathrm{SE}(\hat{K}) \approx \sqrt{24/T}$; for the S&P 500: 0.025 and 0.051

The Delta Method and the Moments That Must Exist

- The **delta method** gives the standard error of a function of an estimator

$$\sqrt{T}(\hat{\theta} - \theta) \to N(0, v) \implies \sqrt{T}\left(g(\hat{\theta}) - g(\theta)\right) \to N\left(0, g'(\theta)^2 v\right)$$

  - θ: a parameter, e.g. a moment; $\hat{\theta}$: its estimator; v: the asymptotic variance of $\hat{\theta}$
  - g: a differentiable function, e.g. the ratio of moments that defines $\hat{S}$; g': its derivative
  - the idea: the first-order Taylor expansion $g(\hat{\theta}) \approx g(\theta) + g'(\theta)(\hat{\theta} - \theta)$
- Any i.i.d. distribution: the delta method gives a Normal limit
  - for $\hat{S}$ only if $\mathbb{E}r^6 < \infty$, for $\hat{K}$ only if $\mathbb{E}r^8 < \infty$
  - $\mathbb{E}$: the expectation; $\mathbb{E}r^6 < \infty$: the sixth moment is finite
  - the variance of $\hat{S}$ involves the sixth moment, that of $\hat{K}$ the eighth
  - whether these moments exist is checked empirically in the section on tails

What the Tests Need: ARCH, Volatility Clustering, Tail Index

- **ARCH** (AutoRegressive Conditional Heteroskedasticity): today's variance depends on past returns
- **Volatility clustering**: days with large moves tend to follow days with large moves
- The **tail index** α measures how fast the probability of extreme values decays

$$P(|r_t| > x) \approx c\, x^{-\alpha} \quad \text{for large } x$$

  - x: a positive threshold; $c > 0$: a scale constant
  - a small α means a heavy tail: the probability decays slowly as x grows
  - so $\mathbb{E}|r_t|^p < \infty$ only for $p < \alpha$, where p is the order of the moment
  - the estimates in this chapter give $\alpha \approx 3$

Three Classical Tests: Normality, Autocorrelation, ARCH Effects

- **JB** (Jarque–Bera): $JB = T\left[\hat{S}^2/6 + (\hat{K}-3)^2/24\right]$
  - the distance of skewness and kurtosis from their Normal-distribution values
- **LB** (Ljung–Box): $Q(m) = T(T+2)\sum_{k=1}^{m} \hat{\rho}_k^2/(T-k)$
  - a weighted sum of the first m squared autocorrelations
  - $\hat{\rho}_k$: the sample autocorrelation at lag k, between r_t and r_{t-k}; here $m = 10$ lags
- **ARCH-LM**: TR^2 from the regression of $\hat{\varepsilon}_t^2$ on $\hat{\varepsilon}_{t-1}^2, \ldots, \hat{\varepsilon}_{t-q}^2$
  - LM: the Lagrange multiplier test; $\hat{\varepsilon}_t = r_t - \bar{r}$: the residuals; $q = 5$ lags
  - T counts observations in the auxiliary regression, after losing these lags
  - R^2: the share of the variation in $\hat{\varepsilon}_t^2$ explained by past squared residuals
- Reading: all three statistics are ≥ 0, and large values reject the null H_0
  - reference distributions: χ_2^2, χ_m^2, χ_q^2; χ_n^2: the chi-squared distribution with n degrees of freedom
  - 5% critical values: $\chi_2^2 = 5.99$, $\chi_5^2 = 11.1$, $\chi_{10}^2 = 18.3$

What Each Test Assumes and Where It Fails on Returns

Test	H_0	χ^2 limit needs	Problem here
JB, Jarque & Bera (1980)	i.i.d., Normal distribution	H_0 itself (all moments exist)	with heavy tails, near-certain rejection for large T
LB, Ljung & Box (1978)	$\rho_1 = \cdots = \rho_m = 0$	i.i.d., not only uncorrelated	volatility clustering: over-rejects
ARCH-LM, Engle (1982)	i.i.d., no ARCH effects	under the i.i.d. H_0: $\mathbb{E}r^4 < \infty$ ($\alpha > 4$)	$\alpha \approx 3 < 4$: size unreliable

- Two moment conditions for ARCH-LM, in two different situations
  - under the i.i.d. null: the χ^2_q limit needs a finite variance of r_t^2, i.e. $\mathbb{E}r^4 < \infty$
  - under GARCH-type dependence: the sample ACF of r_t^2 behind the statistic uses a standard Normal limit under the moment condition $\mathbb{E}r^8 < \infty$ ($\alpha > 8$) Davis & Mikosch (1998)
- The question is not how to run the tests
  - but whether their reference distribution holds for returns

Worked Example: JB for the S&P 500

- 5% critical value of χ^2_2: 5.99
- S&P 500, 1990–2026: $T = 9{,}245$, $\hat{S} = -0.364$, $\hat{K} - 3 = 10.88$
  - skewness term $\hat{S}^2/6 = 0.132/6 = 0.022$
  - kurtosis term $10.88^2/24 = 118.4/24 = 4.93$
  - $JB = 9{,}245 \times (0.022 + 4.93) \approx 45{,}800$, against the critical value 5.99
- Almost all of the statistic comes from the kurtosis term
  - the problem is the tails, not the asymmetry

What a JB Rejection Does and Does Not Show

- Under the null all moments exist
  - so the p-value is valid for the **joint** null "i.i.d. and Normal distribution"
  - heavy tails make JB grow with T
  - this is power, not a broken p-value
- A rejection does not say which part fails
  - it may be the marginal distribution or independence
  - tests of the marginal distribution with HAC (heteroskedasticity and autocorrelation consistent) variances:

▸ Appendix

Robust Alternative: Quantile-Based Skewness and Kurtosis

- Two quantile-based measures Kim & White (2004), which need no moments
  - q_p: the quantile of order p of returns, the value below which a share p of days lies; $q_{0.5}$: the median
- **Hinkley skewness** compares the lengths of the two tails around the median

$$\frac{q_{0.95} + q_{0.05} - 2q_{0.5}}{q_{0.95} - q_{0.05}}$$

  - the numerator is $(q_{0.95} - q_{0.5}) - (q_{0.5} - q_{0.05})$: right tail minus left tail
  - it lies in $[-1, 1]$; a negative value indicates a longer left tail
- **Crow–Siddiqui excess kurtosis**, denoted K_{CS}, compares the width of the tails with the width of the centre

$$K_{CS} = \frac{q_{0.975} - q_{0.025}}{q_{0.75} - q_{0.25}} - 2.91$$

  - 2.91: the value of the ratio for the Normal distribution; a positive value indicates heavier tails
  - both measures are 0 for the Normal distribution

Confidence Intervals from a Moving-Block Bootstrap

- 95% confidence intervals (CI) from a **moving-block bootstrap**
  - it resamples blocks of 20 consecutive days (1,000 resamples)
  - the measure is recomputed on each resampled series
  - the blocks keep the dependence within them
- Valid for stationary, weakly dependent returns with a positive density at these quantiles
- A CI that excludes 0 indicates a significant departure from the Normal distribution

Heavy Tails Are Confirmed without Moments

	Moment skew.	Hinkley [95% CI]	Excess kurt.	Crow–Siddiqui excess [95% CI]
S&P 500	−0.36	−0.064 [−0.092, −0.041]	10.9	1.54 [1.29, 1.79]
BET-TR	−1.40	−0.036 [−0.085, 0.025]	17.8	1.06 [0.73, 1.43]
Bitcoin	−0.70	−0.018 [−0.069, 0.032]	12.0	2.25 [1.88, 2.64]
EUR/RON	0.73	0.041 [−0.022, 0.093]	15.1	4.87 [3.74, 6.13]
Gold	−0.23	0.004 [−0.025, 0.030]	9.3	1.47 [1.29, 1.73]

- Quantile (Crow–Siddiqui) excess kurtosis is positive in all five markets
  - no CI contains 0
- Quantile-based negative skewness is established only for the S&P 500
  - BET-TR's moment skewness −1.40 comes from a few days (19 December 2018)

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Summary Statistics: Five Markets

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Observations	9245	2992	4384	5352	9559
Mean (% per day)	0.033	0.077	0.118	0.007	0.025
Std. dev. (% per day)	1.13	0.98	3.49	0.28	1.01
Annualised volatility (%)	18.0	15.5	66.6	4.4	16.3
Skewness	−0.36	−1.40	−0.70	0.73	−0.23
Excess kurtosis	10.9	17.8	12.0	15.1	9.3
Min / max log return (%)	−12.8 / 11.0	−11.9 / 6.7	−46.5 / 22.5	−2.5 / 2.9	−11.1 / 10.4
Jarque–Bera	45805	40433	26482	51575	34732
Ljung–Box $Q(10)$, r_t	91.6	70.0	20.2	221.7	32.1
Ljung–Box $Q(10)$, r_t^2	8018	714	213	2420	1049
ARCH-LM (5 lags)	2229	371	115	871	483

- Descriptive: heavy tails everywhere (excess kurtosis 9.3–17.8)
  - $Q(10)$ on r_t^2 is 10–90 times larger than on r_t
  - the size of returns is far more predictable than their direction
- Inferential: all i.i.d. p-values are below 0.03
  - but they are valid only for i.i.d. returns with the required moments ($\alpha > 4$ for ARCH-LM)

MFM_ch1_summary_statistics

The Student-t Distribution: a Heavy-Tailed Model

- The Student-t distribution, Praetz (1972): a heavy-tailed model for returns, with density

$$f(x) \propto \left(1 + \frac{(x-\mu)^2}{\nu s^2}\right)^{-(\nu+1)/2}$$

- Notation
  - $f(x)$: the density at x; $\propto$: "proportional to", i.e. without the normalising constant
  - μ: the centre of the distribution; $s > 0$: the scale parameter
  - $\nu > 0$: the degrees of freedom; a small ν means heavy tails; as $\nu \to \infty$ we get the Normal distribution
- For a variable X with this distribution
  - tail $P(X > x) \propto x^{-\nu}$: moments exist only of order $< \nu$
  - variance $s^2\nu/(\nu-2)$ for $\nu > 2$
  - excess kurtosis $6/(\nu-4)$ for $\nu > 4$

Extreme Days: Observed, under the Normal Distribution and under the Student-t

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Days beyond $3\hat{\sigma}$ (%)	1.55	1.44	1.82	2.06	1.42
Days beyond $5\hat{\sigma}$ (count)	30	12	10	29	26
Expected under the Normal distribution, $> 5\hat{\sigma}$	0.005	0.002	0.003	0.003	0.005
Student-t degrees of freedom $\hat{\nu}$ (maximum likelihood)	2.8	3.1	2.2	1.2	2.9

- Under the Normal distribution
  - 0.27% of days lie beyond 3σ (observed 5–8 times more)
  - one 5σ day occurs every 6,900 years of trading (observed 10–30)
- $\hat{\nu}$ is fitted by **maximum likelihood** to the whole distribution: it picks the μ, s and ν that make the observed data most probable
  - the many small returns dominate the fit, so a poor fit in the centre moves $\hat{\nu}$
  - EUR/RON: $\hat{\nu} = 1.2$ reflects that 57% of daily changes are below 0.1%, plus rare jumps

The Hill Estimator: the Tail Index from the Largest Observations

- The tail index α of $X = |r_t|$
  - $\alpha > 2$: finite variance
  - $\alpha > 4$: finite kurtosis
  - power laws were first argued by Mandelbrot (1963) and Fama (1965)
- The **Hill estimator**, Hill (1975): estimates α from the k largest observations only
  - order them from largest to smallest: $X_{(1)} \geq X_{(2)} \geq \cdots \geq X_{(n)}$
  - $X_{(i)}$: the i-th largest value; n: the number of observations
  - k: the number of tail observations; $X_{(k+1)}$ acts as the threshold where the tail starts
- The **Pareto tail**, $P(X > x) = (x/u)^{-\alpha}$ for $x \geq u$, is the exact power law; u is the threshold

The Hill Estimator: the Inverse Mean Log Exceedance

- The Hill estimator is the inverse of the mean log exceedance over the threshold

$$\hat{\alpha}_k = \left[\frac{1}{k} \sum_{i=1}^{k} \ln X_{(i)} - \ln X_{(k+1)} \right]^{-1} = \left[\frac{1}{k} \sum_{i=1}^{k} \ln \frac{X_{(i)}}{X_{(k+1)}} \right]^{-1}$$

  - $\ln(X_{(i)}/X_{(k+1)})$: how far the i-th value exceeds the threshold, on a log scale
  - for a Pareto tail these exceedances average $1/\alpha$
  - large exceedances mean a heavy tail, hence a small $\hat{\alpha}_k$
- Standard error $\mathrm{SE}(\hat{\alpha}_k) \approx \hat{\alpha}_k/\sqrt{k}$; 95% CI: $\hat{\alpha}_k \pm 1.96\,\mathrm{SE}$
  - it assumes independent tail observations and a Pareto tail
  - clustering of extremes makes it larger (Appendix)

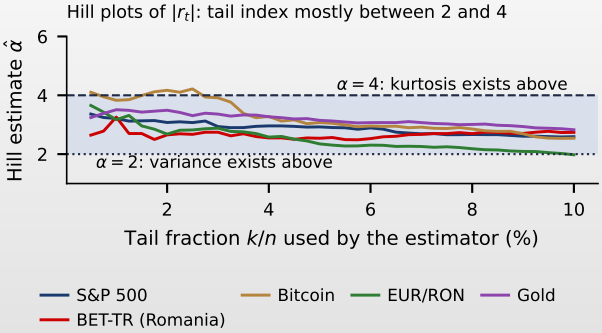
The Hill Estimator by Hand: a Five-Number Example

- Toy data: the five largest $|r_t|$ of a sample, in %: 4.0, 3.0, 2.5, 2.2, 2.0
  - take $k = 4$, so the threshold is $X_{(5)} = 2.0$
- Log exceedances $\ln(X_{(i)}/X_{(5)})$, $i = 1, \ldots, 4$
  - $\ln 2 = 0.693$, $\ln 1.5 = 0.405$, $\ln 1.25 = 0.223$, $\ln 1.1 = 0.095$
  - mean $1.417/4 = 0.354$, so $\hat{\alpha}_4 = 1/0.354 = 2.82$
- SE $\approx 2.82/\sqrt{4} = 1.41$; 95% CI $[0.06, 5.59]$
  - with 4 tail points the estimate says almost nothing; this is why the tables use k in the hundreds
- **Check**: double every value (a market twice as volatile). What happens to $\hat{\alpha}_4$?
  - answer: nothing; the ratios $X_{(i)}/X_{(5)}$ are unchanged, so a change of scale does not move the tail index
- Full worked exercise on market data: Seminar 1, A5

Choosing k: between Variance and Bias

- **Choice of k**
  - small k is noisy
  - large k lets the centre in and biases $\hat{\alpha}$
  - read the level where the **Hill plot** ($\hat{\alpha}_k$ against k) is flat, here $k = 1$–5% of n
- Comparison with the Student-t, whose tail index is $\alpha = \nu$
  - EUR/RON: the Hill estimator gives $\hat{\alpha} = 2.8$ while $\hat{\nu} = 1.2$
  - a Student-t fitted to the whole distribution misdescribes the tails

Hill Plots for Five Markets



- $\hat{\alpha}$ for $|r_t|$ at $k = 2.5\%$ of n
  - S&P 500 3.1, BET-TR 2.7, Bitcoin 4.2, EUR/RON 2.8, gold 3.3
- For $k = 1$–5%: between 2.3 and 3.5 except Bitcoin (3.1–4.2)
- With the same threshold u for both tails, the loss tail is heavier for BET-TR (2.2 versus 3.1) and Bitcoin (2.7 versus 3.4)
- For EUR/RON, the gain tail is heavier
- Tails are modelled with extreme value theory (Chapter 7)

MFM_ch1_heavy_tails

Tail Indices with Confidence Intervals

$\lvert r_t \rvert$, $k = 2.5\%$ of n	n	k	$\hat{\alpha}$	SE i.i.d.	95% CI (i.i.d.)	SE block bootstrap
S&P 500	9245	231	3.07	0.20	[2.68, 3.47]	0.33
BET-TR	2992	74	2.67	0.31	[2.06, 3.28]	0.45
Bitcoin	4384	109	4.22	0.40	[3.43, 5.01]	0.47
EUR/RON	5352	133	2.82	0.24	[2.34, 3.30]	0.28
Gold	9559	238	3.31	0.21	[2.89, 3.73]	0.24

- Moving-block bootstrap (blocks of 20 days, 500 resamples) keeps the clustering of extremes
  - SEs are 1.1–1.7 times the i.i.d. ones

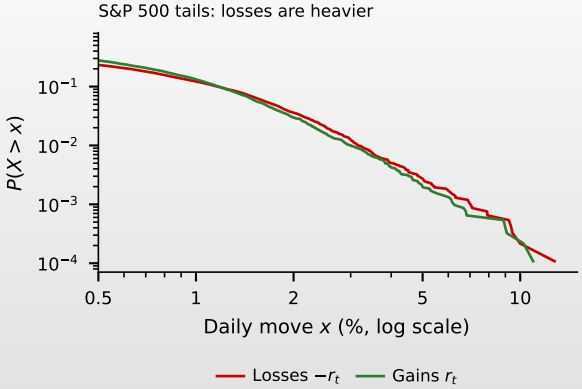
MFM_ch1_inference

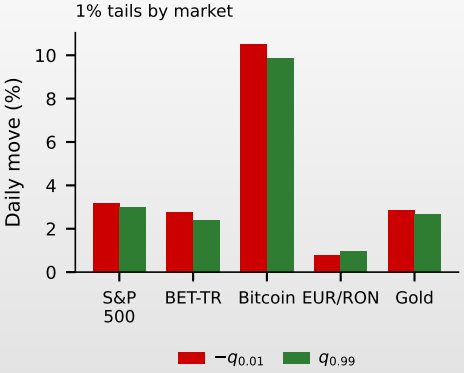
The Fourth Moment Very Likely Does Not Exist

- $\alpha = 4$ is outside every interval except Bitcoin's
  - also when the interval uses the bootstrap SE
  - the fourth moment very likely does not exist
- $\alpha = 2$ is outside every i.i.d. interval
  - with the bootstrap SE only BET-TR reaches it: the lower limit $\hat{\alpha} - 1.96\,\mathrm{SE} = 2.67 - 1.96 \times 0.45 = 1.79$
- Bitcoin's 4.2 uses a threshold where its Hill plot is not yet flat
  - at a common threshold its two tails give 2.71 and 3.36 (Appendix)

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Fact 3: Gain/Loss Asymmetry in the Data





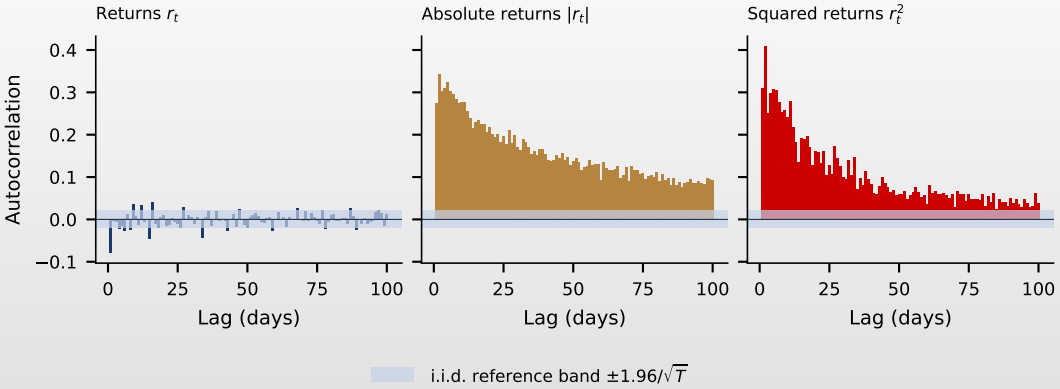
MFM_ch1_heavy_tails

Fact 3: Large Losses Are More Frequent than Large Gains

- Survival functions $P(X > x)$ of losses and gains, log–log scale
  - $P(X > x)$: the share of days on which X exceeds x; $X = -r_t$ for losses, $X = r_t$ for gains
  - large losses are more frequent than gains of the same size
- **VaR** (Value-at-Risk) $1\% = -q_{0.01}$, the daily loss threshold exceeded with probability 1% (Chapter 7)
- VaR 1% exceeds the 99% return quantile $q_{0.99}$ for 4 of 5 markets
  - e.g. for the S&P 500: 3.15% versus 2.97%
- EUR/RON is the exception: leu depreciations (up-moves) are larger
  - $q_{0.99} = 0.94\%$ versus VaR $1\% = 0.76\%$
- A Normal-distribution model with the right variance underestimates such quantiles
  - stress tests based on "3σ events" are too optimistic

MFM_ch1_heavy_tails

Facts 1 and 6: Uncorrelated but Not Independent



MFM_ch1_volatility_clustering

Facts 1 and 6: Direction Does Not Repeat, Size Does

- **Fact 1**: $\text{corr}(r_t, r_{t-k}) \approx 0$, where k is the lag in days
  - S&P 500 1990–2026, lag 1: -0.08
  - i.i.d. reference band: $\pm 1.96/\sqrt{T}$; outside it the autocorrelation is significant at 5%
- **Fact 6, volatility clustering**: $\text{corr}(|r_t|, |r_{t-k}|) > 0$ for many lags (lag 1: 0.27)
  - large moves follow large moves of either sign
- Independence would require zero correlation of **every** function of returns
  - returns are uncorrelated but not independent
- Model $r_t = \sigma_t \varepsilon_t$ with σ_t driven by past returns
  - σ_t: the conditional volatility on day t; ε_t: i.i.d. shocks with mean 0 and variance 1
  - GARCH, generalised ARCH (Chapter 5)

MFM_ch1_volatility_clustering

The ACF under Conditional Heteroskedasticity

- **ACF** (AutoCorrelation Function): the sample autocorrelations $\hat{\rho}_k$, plotted against the lag k

$$\hat{\rho}_k = \frac{\sum_t x_t x_{t-k}}{\sum_t x_t^2}, \qquad x_t = r_t - \bar{r}$$

  - x_t: the demeaned return; $\hat{\rho}_k \in [-1, 1]$
  - $\hat{\rho}_k > 0$: a day above the mean tends to be followed, k days later, by a day above the mean
- Uncorrelated returns whose conditional variance moves: $\hat{\rho}_k \approx N(0, \tau_k)$ Romano & Thombs (1996)
  - τ_k: the variance of $\hat{\rho}_k$, estimated by $\hat{\tau}_k = \sum_t x_t^2 x_{t-k}^2 / (\sum_t x_t^2)^2$
  - i.i.d.: $T\tau_k = 1$, which gives the classical band $\pm 1.96/\sqrt{T}$
  - volatility clustering: $T\tau_k > 1$; the classical band is too narrow and flags autocorrelations as significant too often

The Robust Portmanteau Test $\tilde{Q}(m)$

- Robust t-statistic: $\hat{\rho}_k / \sqrt{\hat{\tau}_k}$, compared with ± 1.96
- Robust portmanteau test Lobato, Nankervis & Savin (2001): null hypothesis $\rho_1 = \cdots = \rho_m = 0$

$$\tilde{Q}(m) = \sum_{k=1}^{m} \frac{\hat{\rho}_k^2}{\hat{\tau}_k}$$

  - the sum of the squared robust t-statistics for the first m lags, so $\tilde{Q}(m) \geq 0$
  - reference distribution χ^2_m; large values reject the null
  - its assumptions: ▸ Appendix
- The difference from the Ljung–Box $Q(m)$
  - $Q(m)$ divides each $\hat{\rho}_k^2$, approximately, by the i.i.d. variance $1/T$
  - $\tilde{Q}(m)$ divides by the estimated variance $\hat{\tau}_k$

With Robust Standard Errors, Only the S&P 500 and EUR/RON Stay Significant

Lag 1	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
$\hat{\rho}_1$	−0.079	0.061	−0.022	0.170	−0.011
$T\hat{\tau}_1$	5.00	4.35	2.89	6.85	2.43
t-statistic, i.i.d. / robust	−7.6 / −3.4	3.3 / 1.6	−1.5 / −0.9	12.4 / 4.7	−1.1 / −0.7

- S&P 500: $Q(10) = 91.6$ but $\tilde{Q}(10) = 19.0$ (p $= 0.041$)
- With robust SEs only the S&P 500 ($t = -3.4$) and EUR/RON ($t = 4.7$) keep a significant lag-1 autocorrelation
  - it is economically small

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Heavy Tails from Random Volatility: Mixtures of Normal Distributions

- The model: the return is a Normal shock times a random volatility

$$r_t = \sigma_t \varepsilon_t, \qquad \varepsilon_t \sim N(0,1) \text{ independent of } \sigma_t$$

- The kurtosis of returns is at least 3

$$K = \frac{\mathbb{E}[r_t^4]}{\mathbb{E}[r_t^2]^2} = 3\,\frac{\mathbb{E}[\sigma_t^4]}{\mathbb{E}[\sigma_t^2]^2} \;\geq\; 3$$

  - the middle equality: $\mathbb{E}[\varepsilon_t^4] = 3$ and $\mathbb{E}[\varepsilon_t^2] = 1$, and σ_t and ε_t are independent
  - the inequality is Jensen's inequality ($\mathbb{E}[X^2] \geq \mathbb{E}[X]^2$) with $X = \sigma_t^2$
  - equality only if σ_t is constant; any variation in volatility creates excess kurtosis

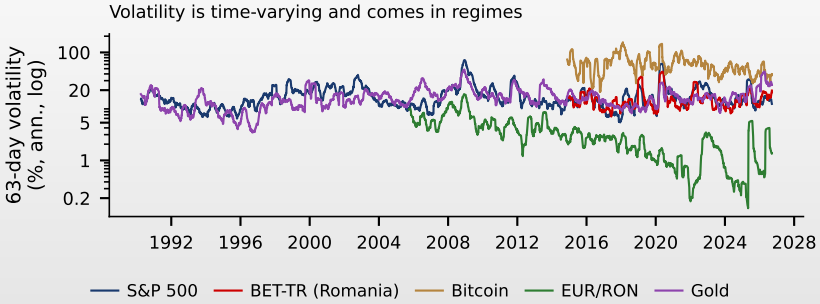
A Numerical Example: Kurtosis 8.33 from Normal Shocks

- Example: $\sigma = 1\%$ on 90% of days and 3% on 10% of days

$$K = 3 \cdot \frac{0.9 \cdot 1 + 0.1 \cdot 81}{(0.9 + 0.1 \cdot 9)^2} = 3 \cdot \frac{9.0}{3.24} = 8.33$$

  - numerator: $\mathbb{E}[\sigma^4] = 0.9 \cdot 1^4 + 0.1 \cdot 3^4$, in units of $(1\%)^4$
  - denominator: $\mathbb{E}[\sigma^2]^2$, with $\mathbb{E}[\sigma^2] = 0.9 \cdot 1^2 + 0.1 \cdot 3^2 = 1.8$
  - the excess kurtosis is 5.33, although each day's return, given σ, is Normally distributed
- This links heavy tails (Fact 2) to volatility clustering (Fact 6)
  - a mixture drawn independently every day has heavy tails but no clustering (Seminar 1, A4 and A9)

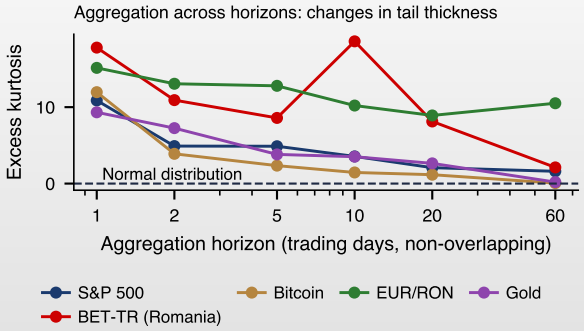
Fact 5: Intermittency in the Data



- Bursts of high volatility separated by calm periods: constant-volatility models cannot produce them
- 63-day rolling volatility $\sqrt{A}\times$ the standard deviation over the window, log scale
  - common spikes: 2008–2009, March 2020
- EUR/RON volatility collapsed after June 2013, under the BNR's managed float

MFM_ch1_volatility_clustering

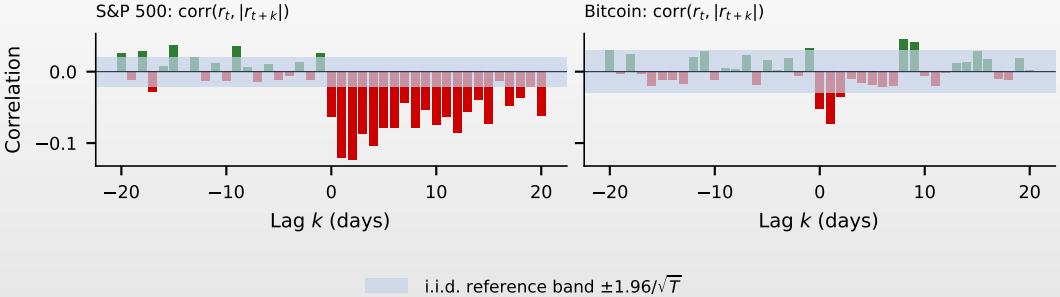
Fact 4: Aggregational Gaussianity



- h-day return: $r_t(h) = r_t + \cdots + r_{t-h+1}$
  - computed in non-overlapping blocks
  - the CLT pulls sums towards the Normal distribution
  - slowly, because daily returns are dependent
- Excess kurtosis: S&P 500 10.9 $\to$ 1.6 at $h = 60$, Bitcoin 12.0 $\to$ 0.1
- EUR/RON stays at an excess kurtosis of about 10 at $h = 60$: aggregation does not bring it close to the Normal distribution
- At $h = 60$ only 154 and 73 returns remain
  - SE of excess kurtosis (i.i.d., Normal): $\sqrt{24/n} = 0.39$ and 0.57
  - n: the number of h-day returns
  - so these values are very imprecise
- $h = 60$ is a quarter for the S&P 500 but two months for Bitcoin

MFM_ch1_heavy_tails

Fact 9: The Leverage Effect



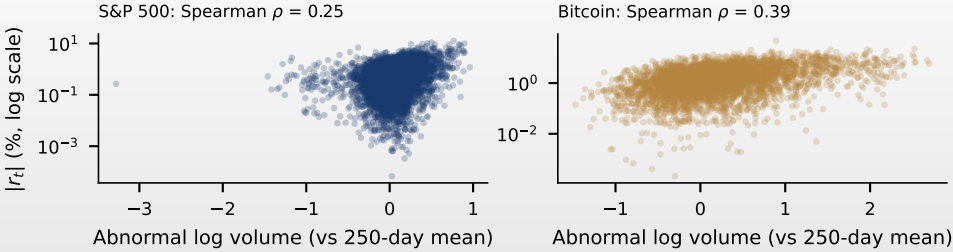
MFM_ch1_leverage_volume

Fact 9: Volatility Rises after Falls

- $L(k) = \text{corr}(r_t, |r_{t+k}|)$, $k > 0$: the correlation of today's return with the size of the return k days later
  - $L(k) < 0$: after falls, returns become larger in absolute value, i.e. volatility rises
- S&P 500: $L(1) = -0.12$, still -0.07 at $k = 10$
  - no effect for $k < 0$
- BET-TR -0.12; weak for Bitcoin (-0.07)
  - negligible for gold (-0.02), reversed for EUR/RON ($+0.10$)
- Possible explanations
  - financial leverage of firms Christie (1982)
  - the retarded volatility model Bouchaud, Matacz & Potters (2001)
- Asymmetric GARCH models: GJR-GARCH (Glosten–Jagannathan–Runkle) and EGARCH (exponential GARCH) (Chapter 5)

MFM_ch1_leverage_volume

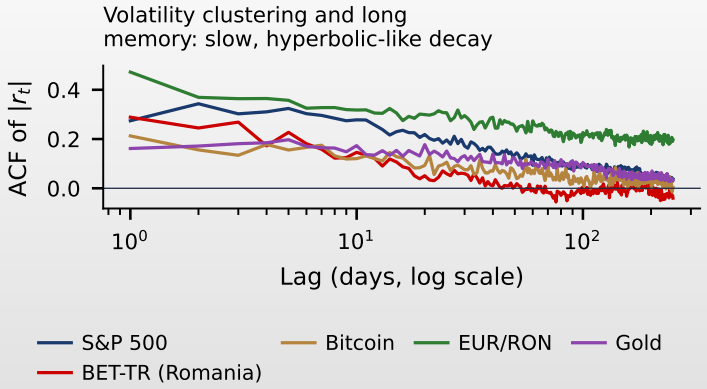
Fact 10: Volume and Volatility Move Together



- Abnormal log volume ($\ln V_t$ minus its 250-day moving average) against $|r_t|$; V_t: the volume traded on day t
  - Spearman rank correlation, the correlation between the ranks of the two series, in $[-1, 1]$: 0.25 (S&P 500) and 0.39 (Bitcoin)
- Mixture-of-distributions hypothesis Clark (1973): information flow drives both price moves and volume
  - survey of the empirical evidence: Karpoff (1987)
- Bitcoin volume is exchange-reported
  - it is partly **wash trading** (an exchange trading with itself) Cong, Li, Tang & Yang (2023)

MFM_ch1_leverage_volume

Fact 8: Slow Decay of the ACF of $|r_t|$



MFM_ch1_volatility_clustering

Fact 8: Exponential or Hyperbolic Decay?

- **Short memory**: $\sum_k |\rho_k| < \infty$
  - ρ_k: the autocorrelation of $|r_t|$ at lag k
  - $c > 0$: a constant
  - e.g. exponential decay $\rho_k \approx c\,\phi^k$, with $0 < \phi < 1$ (GARCH(1,1), Chapter 5)
- **Long memory**: hyperbolic decay $\rho_k \approx c\,k^{2d-1}$, $0 < d < \frac{1}{2}$ Ding, Granger & Engle (1993)
  - d: the memory parameter; a larger d means slower decay, and $\sum_k \rho_k = \infty$
- S&P 500: 0.27 at lag 1, 0.22 at lag 20, still 0.09 at lag 100
  - BET-TR decays fastest
- Slow empirical decay is not yet established long memory
  - structural breaks, i.e. abrupt shifts in the level of volatility, can imitate it

MFM_ch1_volatility_clustering

Estimating the Memory Parameter d

- Long memory in the frequency domain: the spectral density grows without bound at low frequencies

$$f(\lambda) \approx G\lambda^{-2d} \quad \text{as } \lambda \to 0$$

  - $f(\lambda)$: the spectral density, the contribution of cycles of frequency λ to the variance of the series; small λ: long cycles
  - $G > 0$: a constant; d: the memory parameter
  - semiparametric estimators use only the m lowest Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \ldots, m$

- Two estimators (formulas in the Appendix)
  - local Whittle Robinson (1995): $\sqrt{m}(\hat{d} - d) \to N(0, \frac{1}{4})$, so SE $= 1/(2\sqrt{m})$
  - log-periodogram regression Geweke & Porter-Hudak (1983): SE $= \pi/\sqrt{24m}$
  - here $m = T^{0.65}$

$\hat{d} \approx 0.35$–0.50 in All Five Markets

$\lvert r_t \rvert$, $m = T^{0.65}$	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
m	378	181	232	265	386
Local Whittle $\hat{d}$ (SE)	0.48 (0.03)	0.37 (0.04)	0.35 (0.03)	0.39 (0.03)	0.40 (0.03)
Log-periodogram $\hat{d}$ (SE)	0.50 (0.03)	0.41 (0.05)	0.35 (0.04)	0.38 (0.04)	0.42 (0.03)

- $\hat{d} \approx 0.35$–0.50 everywhere
  - at least 8 SEs from $d = 0$ (short memory) with both estimators
- The S&P 500 is close to the stationarity boundary $\frac{1}{2}$
  - for $d \geq \frac{1}{2}$ the series is no longer stationary

MFM_ch1_inference

Long Memory or Structural Breaks?

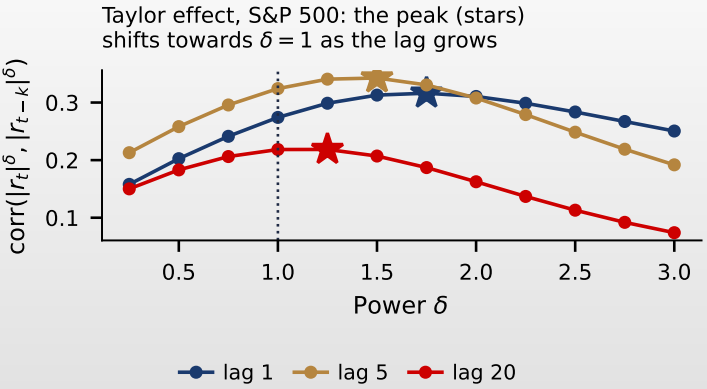
- Rare shifts in the level of volatility produce a slowly decaying ACF of $|r_t|$
  - and $\hat{d} > 0$ without long memory Diebold & Inoue (2001), Mikosch & Stărică (2004)
- Check: locate one variance break with the cumulative sum of squares Inclán & Tiao (1994)
  - then re-estimate d (local Whittle) on each side

$\lvert r_t \rvert$	Break date	Volatility before / after (%)	$\hat{d}$ before (SE)	$\hat{d}$ after (SE)
S&P 500	20 Jul 1998	12.6 / 19.3	0.28 (0.04)	0.51 (0.03)
BET-TR	17 Dec 2018	12.5 / 16.9	0.19 (0.05)	0.37 (0.04)
Bitcoin	19 Jun 2022	74.9 / 47.7	0.39 (0.04)	0.15 (0.05)
EUR/RON	27 Jun 2013	6.6 / 2.2	0.35 (0.04)	0.35 (0.04)
Gold	29 Nov 2005	13.2 / 18.3	0.30 (0.03)	0.37 (0.03)

- $\hat{d}$ stays at least 3 SEs above 0 in every sub-sample (weakest: Bitcoin after 2022, $0.15/0.05 = 3$): one break does not explain the memory
- S&P 500 after 1998: $\hat{d} = 0.51$, above the stationarity boundary $\frac{1}{2}$ but within one SE of it; interpret with care
- $\hat{d}$ moves (Bitcoin $0.39 \to 0.15$): several breaks or a time-varying d remain open; the break dates are descriptive

MFM_ch1_inference

The Taylor Effect



MFM_ch1_volatility_clustering

The Taylor Effect: Persistence Peaks below the Square

- **Taylor effect**: absolute returns $|r_t|$ are more strongly autocorrelated than squared returns r_t^2
  - in the original samples $\text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$ peaks at a power δ close to 1, not at $\delta = 2$
  - documented by Taylor (1986), named by Granger & Ding (1995)
- The autocorrelation $\text{corr}(|r_t|^\delta, |r_{t-k}|^\delta)$ as a function of $\delta > 0$: $\delta = 1$ absolute returns (dotted line), $\delta = 2$ squares
- Our S&P 500 sample: maximum at $\delta = 1.75$ (lag 1), 1.5 (lag 5) and 1.25 (lag 20)
  - the peak is below 2 at every lag and moves towards 1 as the lag grows
  - at lag 1 the peak is closer to 2 than to 1: here the evidence for "$\delta \approx 1$" is mixed at short lags
- The effect motivates
  - power-GARCH models, which model σ_t^δ with an estimated δ (Chapter 5)
  - absolute-value volatility measures

MFM_ch1_volatility_clustering

Which Inference Survives When Moments Are Missing

Statistic	Needs	With Hill $\hat{\alpha} \approx 2.3$–3.5 (Bitcoin: 4.2)
Mean, variance of r_t	$\alpha > 2$	finite, slow convergence
ACF of $\lvert r_t \rvert$, robust ACF bands, $\bar{Q}$	$\alpha > 4$ for Normal limits	doubtful
Kurtosis, ACF of r_t^2, ARCH-LM, Q on r_t^2	$\alpha > 4$ to exist (and for ARCH-LM's χ^2 under i.i.d.), $\alpha > 8$ for Normal limits under GARCH-type dependence	no standard inference
Skewness	$\alpha > 6$ for a Normal limit	no standard inference
Quantiles, number of tail days, Hill $\hat{\alpha}$	no moments	valid, with dependence-robust SEs

- With an infinite fourth moment the sample ACF of r_t^2 converges slowly
  - its limit is not the Normal distribution Davis & Mikosch (1998)
  - the $\pm 1.96/\sqrt{T}$ band and the p-values of Q on r_t^2 and ARCH-LM are not reliable
- The conclusion "volatility clusters" is robust
  - the exact magnitudes are not robust
  - one reason behind the Taylor effect: $\lvert r_t \rvert$ needs fewer finite moments than r_t^2

Facts 7 and 11: Conditional Tails and Time Scales

Fact 7: conditional heavy tails

- Fit a volatility model, e.g. GARCH (Chapter 5)
  - and standardise: $\hat{z}_t = r_t / \hat{\sigma}_t$, with $\hat{\sigma}_t$ the model's volatility estimate for day t
- The kurtosis of $\hat{z}_t$ is lower than that of r_t, still above 3
- Models need heavy-tailed innovations (Student-t, possibly skewed)

Fact 11: asymmetry in time scales

- Weekly volatility predicts daily volatility better than the reverse
- Traders with different horizons react at different frequencies
- The heterogeneous autoregressive model of realised volatility (**HAR-RV**, Chapter 9)
  - realised volatility is the sum of squared intraday returns

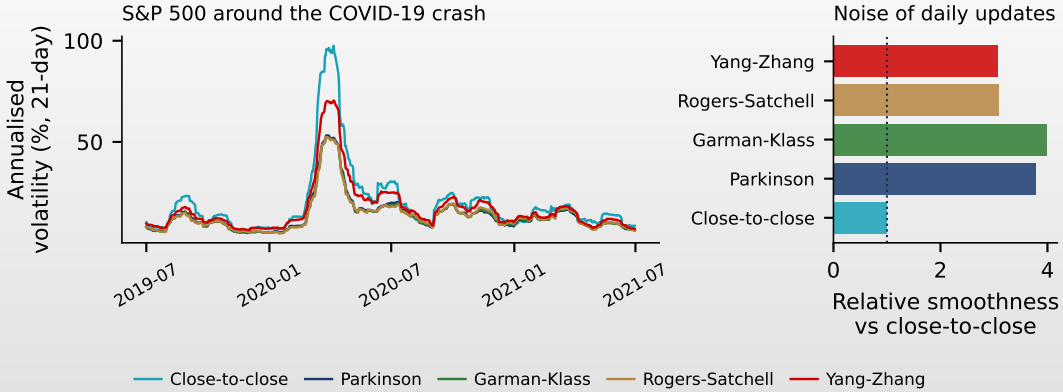
- Both facts are documented in Cont (2001)
- Returns standardised by realised volatility from intraday data are close to the Normal distribution
  - Andersen, Bollerslev, Diebold & Ebens (2001) (Chapter 9)

Differences between the Five Markets

Market	Institutional feature	What differs	Candidate explanation
Bitcoin	trades 24/7; $A = 365$	volatility 66.6% versus 18.0% (S&P 500); worst day -46.5%; weak leverage (-0.07); ACF of $\lvert r_t \rvert$ at lag 100: 0.02 versus 0.09	no corporate leverage; volatility rises after large moves of either sign
EUR/RON	BNR daily fixing, managed float	volatility 4.4% but excess kurtosis 15.1; $\hat{\rho}_{\mathbf{1}} = +0.17$; largest moves October 2008 ($+2.9\%$, -2.5%)	adjustments smoothed over several days
BET-TR	total return, frontier/emerging market	skewness -1.40; worst day -11.9% on 19 Dec 2018 (bank-asset tax announced on 18 Dec, Government Emergency Ordinance 114/2018); $\hat{\rho}_{\mathbf{1}} = +0.06$	policy shocks; **non-synchronous trading** of constituents Campbell, Lo & MacKinlay (1997), ch. 3

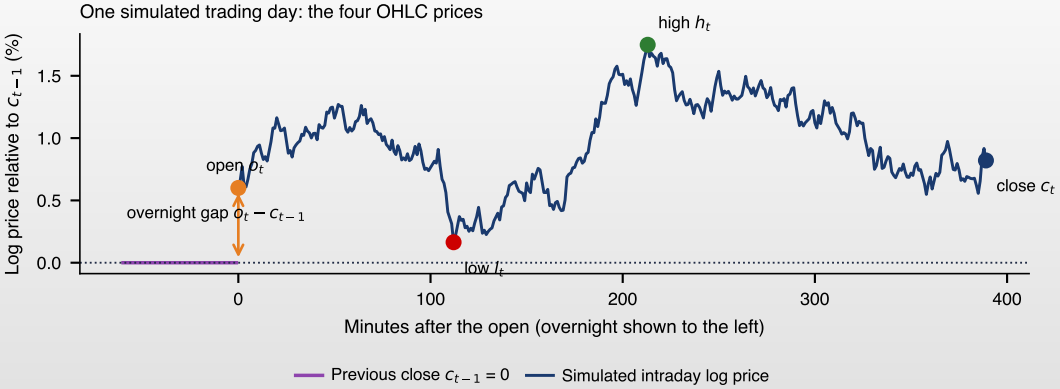
- The explanations are hypotheses, not identified causes
- Low volatility is not low tail risk
- Heavy tails and volatility clustering hold in all five markets
  - autocorrelation and asymmetry depend on institutions

Volatility Estimators on the S&P 500



MFM_ch1_vol_estimators

Open, High, Low and Close (OHLC) Prices of One Day



MFM_ch1_seminar_tests

What the Day's High and Low Add

- **OHLC**: four prices of day t
  - O_t: the open
  - H_t: the high
  - L_t: the low
  - C_t: the close
- The chart: one simulated day; the vertical axis is the log price relative to the previous close c_{t-1}, in %
  - the overnight gap $o_t - c_{t-1}$ sits left of the open
- **Range-based** estimators use the whole path of the day
  - the range H_t/L_t is large on volatile days
  - even if the close equals the open

MFM_ch1_seminar_tests

The Log Range and the Efficiency of a Volatility Estimator

- Logs of the four prices

$$o_t = \ln O_t, \qquad h_t = \ln H_t, \qquad l_t = \ln L_t, \qquad c_t = \ln C_t$$

  - the lower-case letter is the log of the price with the same initial
- Log range of day t

$$h_t - l_t = \ln(H_t/L_t) \geq 0$$

  - approximately the percentage gap between the day's high and low
- Three log returns of day t
  - close-to-close: $c_t - c_{t-1}$
  - overnight: $o_t - c_{t-1}$
  - intraday: $c_t - o_t$
- **Close-to-close estimator**: the standard deviation of $c_t - c_{t-1}$
  - one observation per day
- **Efficiency** of an estimator X: $\text{Var}(\hat{\sigma}^2_{CC})/\text{Var}(\hat{\sigma}^2_X)$
  - an efficiency of 5 means the same precision with 5 times fewer days

The Parkinson Estimator: Variance from the Day's Range

- **Parkinson**, Parkinson (1980): estimates the daily variance from the daily range only

$$\hat{\sigma}_P^2 = \frac{1}{4 \ln 2} \frac{1}{n} \sum_{t=1}^{n} (h_t - l_t)^2$$

  - n: the number of days; $h_t - l_t$: the log range of day t
  - for a Brownian motion with daily variance σ^2, $\mathbb{E}[(h_t - l_t)^2] = 4 \ln 2 \, \sigma^2$; dividing by $4 \ln 2$ makes the estimator unbiased
  - $\hat{\sigma}_P^2$ is a daily variance; the annual volatility is $\sqrt{A \, \hat{\sigma}_P^2}$

- Assumption: a Brownian motion without drift and without overnight jumps
  - Brownian motion: the continuous-time random walk (Chapter 11)
  - efficiency relative to close-to-close: about 5 (Seminar 1, A8)

The Garman–Klass Estimator: the Range plus the Open-to-Close Move

- **Garman–Klass**, Garman & Klass (1980): combines the range with the open-to-close move

$$\hat{\sigma}^2_{GK} = \frac{1}{n}\sum_{t=1}^{n}\left[\tfrac{1}{2}(h_t - l_t)^2 - (2\ln 2 - 1)(c_t - o_t)^2\right]$$

  - $c_t - o_t$: the open-to-close log return
  - the weights $\frac{1}{2}$ and $2\ln 2 - 1 \approx 0.386$ come from the minimum-variance estimator derived by Garman and Klass
  - the same assumptions as Parkinson: no drift, no overnight jumps
- Efficiency relative to close-to-close: 7–8

The Rogers–Satchell Estimator: Unbiased under a Drift

- **Rogers–Satchell**, Rogers & Satchell (1991): unbiased under a non-zero **drift** (expected trend of the log price)

$$\hat{\sigma}^2_{RS} = \frac{1}{n}\sum_{t=1}^{n}[(h_t - o_t)(h_t - c_t) + (l_t - o_t)(l_t - c_t)]$$

  - the terms measure how far the high and the low moved away from the open and from the close
  - both products are ≥ 0, because $l_t \leq o_t, c_t \leq h_t$
  - for a Brownian motion with drift their mean is exactly the daily variance
  - it does not include the overnight jump

The Yang–Zhang Estimator: Adding the Overnight Jump

- **Yang–Zhang**, Yang & Zhang (2000): adds the overnight (close-to-open) jump

$$\hat{\sigma}^2_{YZ} = \hat{\sigma}^2_{\text{overnight}} + k\,\hat{\sigma}^2_{\text{open-close}} + (1-k)\,\hat{\sigma}^2_{RS}, \quad k = \frac{0.34}{1.34 + \frac{n+1}{n-1}}$$

  - $\hat{\sigma}^2_{\text{overnight}}$: the sample variance of $o_t - c_{t-1}$, the overnight return
  - $\hat{\sigma}^2_{\text{open-close}}$: the sample variance of $c_t - o_t$
  - k: the weight, chosen to make the combination minimum-variance (not the Hill k); $k \approx 0.145$ for large n
  - minimum-variance combination among drift-independent estimators robust to opening jumps
- Range estimators are the bridge to realised volatility from intraday data (Chapter 9)

Missing the Overnight Jumps, Range Estimators Read Lower

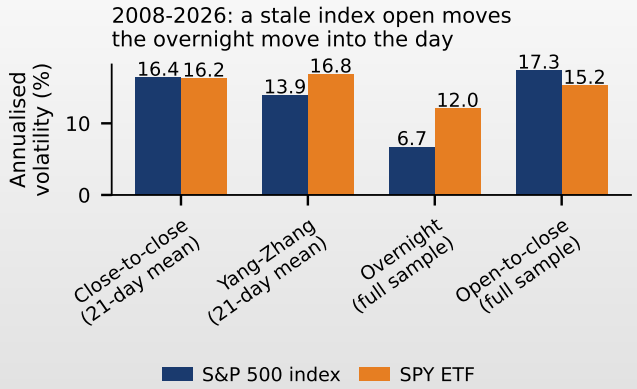
- 21-day windows, 2008–2026; averages
  - close-to-close 16.4%, Yang–Zhang 13.9%, Parkinson 13.0%, Garman–Klass 12.4%, Rogers–Satchell 12.3%
- Pure range estimators are lower: they miss overnight gaps
- Right panel, "relative smoothness" of estimator X

$$\left[\frac{\text{std. dev. of daily changes, close-to-close estimate}}{\text{std. dev. of daily changes, estimate } X}\right]^2$$

  - both estimates use 21-day windows
  - equal to 1 for close-to-close; larger values mean the estimate jumps less from one day to the next
- The range-based series are 3–4 times smoother on this measure
  - a property of the estimates, not by itself proof of efficiency: it does not compare them with the true σ

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Stale Opens: the S&P 500 Index versus SPY, an Exchange-Traded Fund (ETF)



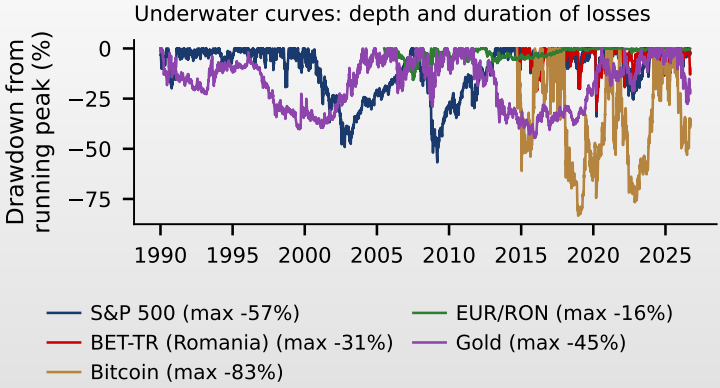
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Stale Opens: Why the Index Open Lags Behind

- **Stale open**: the index open is computed from the constituents' first prices
  - at 9:30 many stocks have not traded yet
  - so their previous close enters the open
- SPY, the ETF that tracks the S&P 500, opens at a traded price
- Open = previous close
  - about 96% of days in 1996–2005 and still 16% in 2007
  - 3.5% over 2008–2026 (up to 26% in 2012–2013)
  - range-based estimators therefore use data from 2008 on
- A stale open moves part of the overnight move into the day
  - index overnight volatility: 6.7%
  - SPY, whose open is a traded price: 12.0%
- Yang–Zhang falls below close-to-close on the index only
  - use a traded instrument for estimators that use the open (Seminar 1, B15)

MFM_ch1_seminar_tests

Drawdowns Across Markets



MFM_ch1_risk_measures

The Maximum Drawdown: How Far a Price Can Fall from a Peak

- Drawdown $DD_t = P_t / \max_{s \le t} P_s - 1$: the loss from the peak reached up to day t (Chapter 0)
  - $\max_{s \le t} P_s$: the highest price up to day t; $DD_t \in [-1, 0]$, with 0 at a new peak
  - maximum drawdown $MDD = \min_t DD_t$: the deepest fall from a peak
- MDD in the five markets
  - Bitcoin -83%, S&P 500 -57%, gold -45%
  - BET-TR -31% (March 2020), EUR/RON -16%
- S&P 500: 9 October 2007 (1,565.15) to 9 March 2009 (676.53): -56.8%
  - **recovery**: the price regaining its previous peak
  - after a fall of x it needs a gain of $1/(1-x) - 1$
  - here $1/(1 - 0.568) - 1 = +131\%$
- The MDD grows with the sample length
  - compare only equal horizons (Appendix)

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Recovery Duration after the Maximum Drawdown

- **Recovery duration**: the time from the peak until the price reaches it again (recovery, defined on the previous slide)

	MDD	Peak	Trough	Peak regained
S&P 500	−57%	Oct 2007	Mar 2009	Mar 2013 (5.5 years)
BET-TR	−31%	Jan 2020	Mar 2020	Dec 2020 (0.9 years)
Bitcoin	−83%	Dec 2017	Dec 2018	Nov 2020 (3.0 years)
EUR/RON	−16%	Dec 2005	Jul 2007	Jan 2008 (2.0 years)
Gold	−45%	Sep 2011	Dec 2015	Jul 2020 (8.9 years)

- Depth and duration are different risks
  - gold's drawdown was shallower than Bitcoin's but lasted three times longer
- Fast V-shaped recoveries (BET-TR 2020)
  - versus long underwater periods (gold 2011–2020)
- For EUR/RON a "drawdown" is a fall of the euro against the leu

MFM_ch1_risk_measures

The Sharpe Ratio: Excess Return per Unit of Volatility

- **Sharpe ratio** Sharpe (1966), Sharpe (1994)
  - the excess return earned per unit of volatility

$$SR = \frac{\mu_{\text{ann}} - r_f}{\sigma_{\text{ann}}}$$

  - r_f: the risk-free rate; $r_f = 0$ here, for comparability
  - $\mu_{\text{ann}} = A\bar{R}$: the annual mean of **simple** returns
  - $\bar{R}$: the mean daily simple return
  - σ_{ann}: the annual volatility of simple returns
  - a higher SR means more return for the same risk
- S&P 500 1990–2026: $\mu_{\text{ann}} = 9.95\%$, $\sigma_{\text{ann}} = 17.97\%$, $SR = 0.55$
  - μ_{ann} = mean log return 8.33% plus $\approx \sigma^2/2 = 1.62$ pp
  - this is not the log-return mean $A\bar{r}$ of the annualisation slide: with 8.33% one would get $8.33/17.97 = 0.46$

The Sortino Ratio: Only Losses in the Denominator

- **Sortino ratio** Sortino & van der Meer (1991)
  - the same numerator as the Sharpe ratio
  - in the denominator: only the volatility of losses

$$SoR = \frac{\mu_{\text{ann}} - r_f}{\sigma^-_{\text{ann}}}, \qquad \sigma^- = \sqrt{\frac{1}{T}\sum_{t=1}^{T} \min(R_t, 0)^2}$$

  - σ^-: the daily downside deviation (in the seminar: the downside semi-deviation $\hat{\sigma}_D$)
  - $\min(R_t, 0)$ keeps only negative returns
  - the average runs over all T days; positive days count as 0
  - $\sigma^-_{\text{ann}} = \sqrt{A}\,\sigma^-$: the annualised value
- S&P 500: $\sigma^-_{\text{ann}} = 12.68\%$, $SoR = 0.78$

The Calmar Ratio: Growth per Unit of Maximum Drawdown

- **Calmar ratio**
  - the annual growth earned per unit of maximum drawdown

$$\text{Calmar} = \frac{\text{CAGR}}{|MDD|}$$

  - CAGR: the compound annual growth rate
  - MDD: the maximum drawdown in the sample
- S&P 500: $8.69\%/56.78\% = 0.15$
  - the value is driven by one episode
- The Calmar ratio has no usable sampling theory
  - its maximum drawdown grows with the sample length

Sharpe Ratios with Standard Errors

- Delta method, i.i.d. returns from the Normal distribution Lo (2002)
  - SR_d: the daily Sharpe ratio; $\widehat{SR}_d$: its estimate; $\mathrm{Var}(\widehat{SR}_d) \approx (1 + SR_d^2/2)/T$
  - annualised, $SR_{\text{ann}} = \sqrt{A}\, SR_d$: $\mathrm{SE}(\widehat{SR}_{\text{ann}}) \approx \sqrt{A(1 + SR_d^2/2)/T} \approx 1/\sqrt{Y}$, with $Y = T/A$ years
  - the span matters, not the frequency

	S&P 500	BET-TR	Bitcoin	EUR/RON	Gold
Years Y	36.7	12.0	12.0	21.2	36.7
Sharpe ratio (SE, Normal)	0.55 (0.17)	1.34 (0.29)	0.98 (0.29)	0.43 (0.22)	0.48 (0.17)

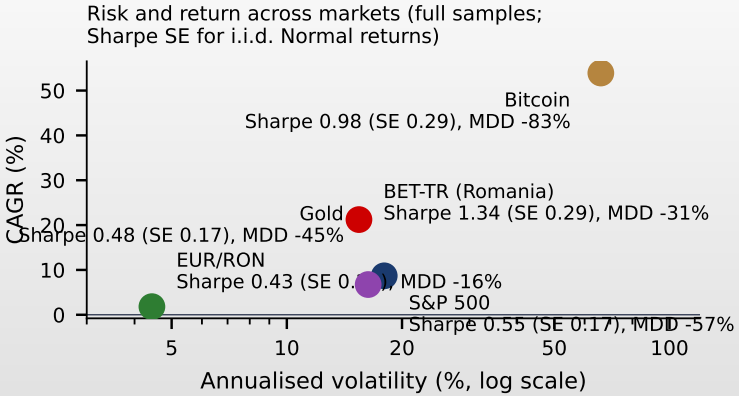
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BET-TR versus S&P 500: the Sharpe Difference Is Not Significant at 5%

- Clustering: HAC or studentised block-bootstrap tests Ledoit & Wolf (2008)
  - they still need finite fourth moments, which the Hill estimates put in doubt
- BET-TR versus S&P 500, common days 2014–2026
  - difference 0.61, $z = 1.75$ ($p = 0.08$)
  - z: the difference divided by its standard error, compared with $N(0, 1)$
  - block-bootstrap 95% CI $[-0.01, 1.21]$ (Seminar 1, B14)

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Risk and Return Across Markets



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Why the Risk–Return Chart Is Not a Ranking

- Labels: the Sharpe ratio and, in brackets, its standard error (SE)
  - SE: the formula of Sharpe ratios with standard errors, valid for i.i.d. Normal simple returns
  - full table in the ▸ Appendix
- Different sample periods: not a ranking
- S&P 500 without dividends; BET-TR with dividends and in RON
  - RON short-term rates were well above USD rates
  - with $r_f = 0$ the BET-TR Sharpe ratio is inflated
- Heavy tails and negative skewness flatter the Sharpe ratio
  - report it with drawdowns
  - and with tail-risk measures: VaR 1% and the tail index $\hat{\alpha}$

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Case Study (1/11): How Much of Volatility Do Past Returns Alone Explain?

- Paper: Guyon & Lekeufack (2023), *Quantitative Finance*
  - the paper builds on Bollerslev, Hood, Huss & Pedersen (2018)
- Research question
  - what share of an equity index's volatility is explained by its past daily returns alone?
  - what functional form does this dependence take?
- Three dependent variables, modelled separately (Sections 3.4–3.5 of the paper)
  - VIX (the Cboe volatility index): the implied volatility of the S&P 500 over the next 30 days, computed from option prices
  - **implied volatility**: the volatility that makes an option-pricing model match observed option prices, i.e. the market's forward-looking volatility (Chapter 12)
  - VIX9D: the same measure over a 9-day horizon
  - next-day realised volatility, computed from 5-minute returns

Case Study (2/11): One Model for Volatility Clustering, Long Memory and Leverage

- Requirement: a single model must reproduce simultaneously
  - volatility clustering (Fact 6)
  - the slow decay of the autocorrelation of $|r_t|$ (Fact 8)
  - the leverage effect (Fact 9)
- Why it matters: if volatility is mostly a function of the price path, it is mostly endogenous
  - a **path-dependent volatility (PDV)** model needs little extra randomness
  - unlike stochastic volatility models (Chapters 11–12)
- Data: daily closes of the S&P 500 (US), Euro Stoxx 50 (euro area), FTSE 100 (UK), DAX (Germany) and Nikkei 225 (Japan) indices
  - training 1 Jan 2000 – 31 Dec 2018
  - test 1 Jan 2019 – 15 May 2022

Case Study (3/11): Volatility as a Function of the Trend and of Recent Volatility

- Equation (3.7) in the paper: today's volatility is a linear function of two summaries of past returns

$$\sigma_t = \beta_0 + \beta_1 R_{1,t} + \beta_2 \Sigma_t$$

- Notation
  - σ_t: the dependent variable, for example the VIX at the close of day t
  - S_{t_i}: the index level at the close of day t_i; $t_i \leq t$ runs over the days up to and including t
  - $r_{t_i} = S_{t_i}/S_{t_{i-1}} - 1$: the **simple** return of day t_i (the paper's notation; elsewhere in this chapter r_t is the log return)
  - $R_{1,t}$: the trend component, built from past returns
  - Σ_t: the volatility component, built from past squared returns
  - $\beta_0, \beta_1, \beta_2$: coefficients estimated from the data

Case Study (4/11): How Much Does an Old Day Count?

- A **kernel** $K(\tau)$ gives the weight of a day of age τ, with $\tau = i\Delta t$ for a day i days old and $\Delta t = 1/252$
- The daily weights $K(i\Delta t)\Delta t$ are positive and sum to 1 (the vertical axis of the kernel chart)
- The paper uses a **time-shifted power law (TSPL)**, eq. (3.9)

$$K(\tau) = \frac{(\tau + \delta)^{-\alpha}}{Z_{\alpha,\delta}}, \qquad \alpha > 1,\ \delta > 0,\ \tau = i\Delta t,\ i = 0, 1, \ldots, 999$$

  - $Z_{\alpha,\delta}$: the normalising constant over the last 1,000 days
  - α: how fast the weights decay; a small α means long memory
  - δ: the time shift, in years; a small δ concentrates the weight on the last few days ($\delta = 0.05$ years ≈ 12.6 days)
- Resulting shape: large weight on the last few days (short memory) and a slowly decaying power-law tail (long memory, Fact 8)
- The model uses two kernels: K_1 for the trend and K_2 for volatility, with parameters (α_1, δ_1) and (α_2, δ_2)
- Here α, δ and K are not the tail index, the Taylor-effect power or the kurtosis

Case Study (5/11): The Recent Trend and a Weighted Historical Volatility

- Trend component: a weighted sum of past returns

$$R_{1,t} = \sum_{t_i \le t} K_1(t - t_i)\, r_{t_i}$$

  - $t - t_i$: the age of day t_i, measured in years as in the paper (one day $= 1/252$ years); $K_1(t - t_i)$: the weight of that day
  - $R_{1,t} < 0$ after a run of falls, $R_{1,t} > 0$ after a run of rises
- Volatility component: the square root of a weighted sum of squared returns

$$\Sigma_t = \Big(\sum_{t_i \le t} K_2(t - t_i)\, r_{t_i}^2 \Big)^{1/2}$$

  - The weights $K_j(i\Delta t)\Delta t$ sum to 1: $R_{1,t}$ and Σ_t are weighted means, annualised with 252 and $\sqrt{252}$
- Interpreting the coefficients
  - $\beta_1 < 0$: volatility rises after falls ($R_{1,t} < 0$), i.e. the leverage effect
  - $\beta_2 > 0$: high recent volatility raises today's volatility, i.e. volatility clustering, as in GARCH
  - $\beta_2 < 1$ is an empirical finding, not a constraint ($\hat{\beta}_2 = 0.87$)
  - the VIX moves less than one-for-one with Σ_t
  - $\beta_0 > 0$ supplies part of the level

Case Study (6/11): Seven Parameters, Scored Also out of Sample

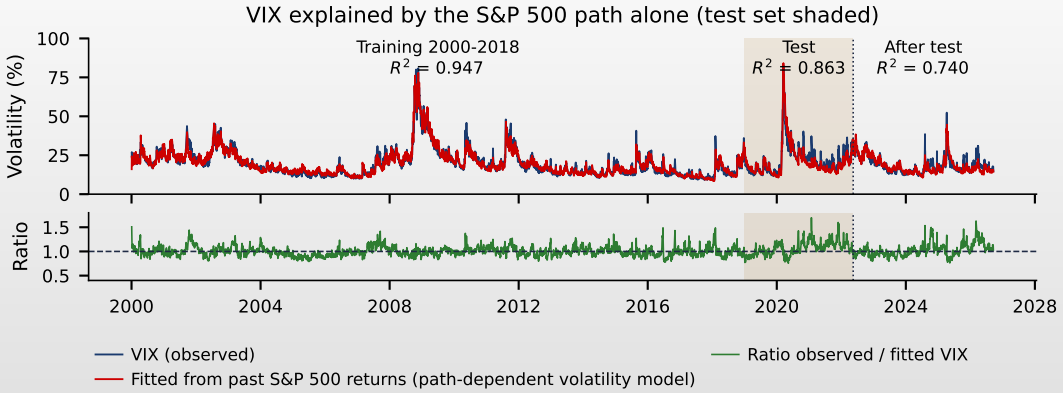
- Seven parameters: $\beta_0, \beta_1, \beta_2, \alpha_1, \delta_1, \alpha_2, \delta_2$
  - estimated by nonlinear least squares on the training set: minimise $\sum_t (\sigma_t - \hat{\sigma}_t)^2$
  - $\hat{\sigma}_t$: the model value for day t
  - estimation details: Appendix
- Scores, computed separately on the training and the test set
  - RMSE (root mean squared error) $= \sqrt{\frac{1}{T}\sum_t (\sigma_t - \hat{\sigma}_t)^2}$: the typical error, in volatility units (decimals: 0.01 = 1 VIX point)
  - $R^2 = 1 - \sum_t (\sigma_t - \hat{\sigma}_t)^2 / \sum_t (\sigma_t - \bar{\sigma})^2$: the share of the variation of σ_t explained by the model
  - $\bar{\sigma}$: the mean of σ_t; $R^2 = 1$ is a perfect fit, $R^2 = 0$ is no better than the mean

Case Study (7/11): Past Returns Explain 80–92% of Implied Volatility

Index	Implied volatility (paper, Table 2) R^2 train	R^2 test	Realised volatility (paper, Table 4) R^2 train	R^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (VIX), our replication	0.947	0.863	–	–
S&P 500 (9-day VIX)	0.876	0.914	–	–
Euro Stoxx 50	0.929	0.913	0.672	0.682
Nikkei 225	0.890	0.800	0.563	0.504

- Past index returns alone explain 80–92% of the variance of implied volatility out of sample
  - FTSE 100 and DAX in the Appendix
- On our data (S&P 500 and VIX): RMSE 0.020 (2 VIX points) training and 0.034 test
- Realised volatility scores are lower
  - 5-minute realised volatility is a noisy target (Remark 3 in the paper)

Case Study (8/11): Observed VIX and VIX Fitted from S&P 500 Returns



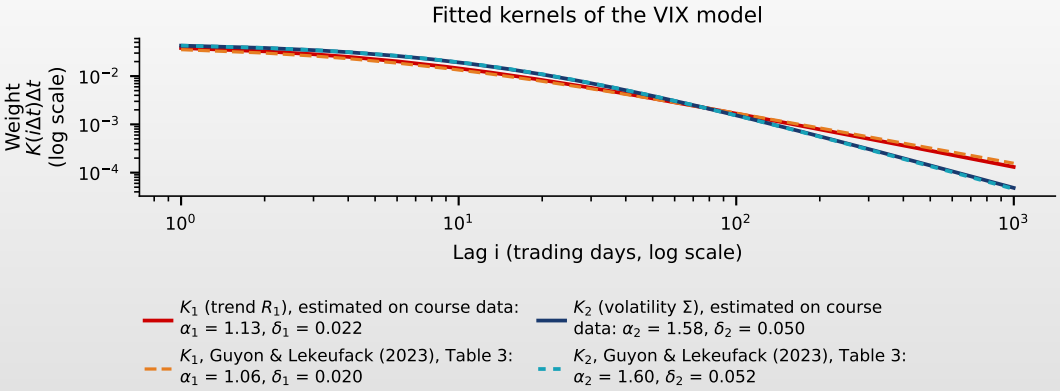
MFM_ch1_path_dependent_vol

Case Study (9/11): After 2022, a Smaller Error but Also a Lower R^2

- Fitted on 3 Jan 2000 – 31 Dec 2018 (4,779 days)
  - then parameters frozen
- The test set is shaded
- After the test set, 16 May 2022 – 18 Sep 2026: $R^2 = 0.740$, RMSE 0.026
  - a lower RMSE, but the VIX varied less (standard deviation 5.0 versus 9.3 points)
  - so R^2 falls
- Ratio observed/fitted
  - mean 0.85 in April–May 2020
  - mean 1.16 from October 2020 to May 2022
  - 8 April 2025: VIX 52.3 versus 39.2 fitted

MFM_ch1_path_dependent_vol

Case Study (10/11): Fitted Kernels



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Case Study (11/11): One Kernel for Short and Long Memory; What Stays Exogenous

- Daily weights $K(i\Delta t)\Delta t$, log–log scale; in parentheses, Table 3 of the paper
  - K_1 (trend): $\hat{\alpha}_1 = 1.13$ (1.06), $\hat{\delta}_1 = 0.022$ (0.020)
  - K_2 (volatility): $\hat{\alpha}_2 = 1.58$ (1.60), $\hat{\delta}_2 = 0.050$ (0.052)
- The last 5 days carry 20% of the weight of K_2
  - the days beyond lag 250 still carry 10%
  - short and long memory in one kernel
- Limits stated by the authors (Section 5 of the paper)
  - a small exogenous part remains
  - the ratio of observed to fitted VIX mean-reverts around 1
  - this ratio needs a stochastic factor
- Open extension (project seed)
  - does the same formula describe the realised volatility of Bitcoin or of BET-TR?

MFM_ch1_path_dependent_vol

From Evidence to Model Requirements

Evidence (five markets)	Model requirement	Where in this course
quantile (Crow–Siddiqui) excess kurtosis > 0 in all five markets; Hill $\hat{\alpha} \approx 2.3$–3.5 (Bitcoin: 4.2)	heavy-tailed innovations, also conditionally	Ch. 5, Ch. 7 (extreme value theory)
ACF of $\lvert r_t \rvert$ positive for months; $\hat{d} \approx 0.35$–0.50	persistent, time-varying volatility; long memory or breaks	Ch. 5, Ch. 9
$L(1) = -0.12$ (S&P 500, BET-TR); reversed for EUR/RON	asymmetric response to bad news where it exists	Ch. 5 (GJR-GARCH, EGARCH)
$\hat{\rho}_1$ significant (robust SEs) only for the S&P 500 and EUR/RON; economically small in all markets	approximately uncorrelated returns; market microstructure matters	Ch. 2, Ch. 10
quantile skewness negative only for the S&P 500	asymmetry is market-specific	Ch. 7–8

- The random walk with i.i.d. increments from the Normal distribution (Bachelier–Black–Scholes world) violates Facts 2, 3 and 5–11
  - Facts 1 and 4 hold trivially
- ARCH-LM significance supports volatility clustering descriptively
  - with doubtful fourth moments it is not a formal proof

Open Question: Are Tail Indices Stable over Time?

- The tail index α is usually treated as a constant
- Crypto-assets changed in structure
  - futures (2017), spot Bitcoin ETFs (January 2024), institutional holders
  - Pele et al. (2023): the tail factor separates cryptos from other asset classes
- Hypothesis proposed by an AI assistant
  - after the launch of the spot ETFs (January 2024), the entry of institutional investors has made the tails of Bitcoin's return distribution thinner
  - i.e. it has raised the tail index α

The Discovery Loop for Bitcoin's Tail Index

- **Literature**: ask for peer-reviewed work with DOIs, then resolve each DOI yourself
  - Hoga (2017): change-point tests for the tail index valid under β-mixing dependence
  - β-mixing: dependence that vanishes with the distance in time, e.g. in GARCH models
- **Competing hypotheses**
  - a change in tail shape
  - a fall in volatility only (scale does not move α)
  - sampling noise
- **Data and code**: replicate the chapter's Hill estimates first (Bitcoin 4.2, S&P 500 3.1)
- Fixed in advance
  - $k = 2.5\%$ of n, the break date 11 January 2024
  - block-bootstrap SEs, a placebo asset

Mini-Case: Did Bitcoin's Tail Get Thinner after the ETFs?

- Test: Hill estimate for $|r_t|$, $k = 2.5\%$ of n
  - $H_0 : \alpha_{\text{post}} = \alpha_{\text{pre}}$ against $H_1 : \alpha_{\text{post}} > \alpha_{\text{pre}}$
  - $\hat{\alpha}_{\text{pre}}$, $\hat{\alpha}_{\text{post}}$: the Hill estimates in the two subperiods; s: their standard errors, from a moving-block bootstrap (blocks of 20 days)
  - $z = (\hat{\alpha}_{\text{post}} - \hat{\alpha}_{\text{pre}})/\sqrt{s_{\text{pre}}^2 + s_{\text{post}}^2}$: the difference divided by its standard error
  - under H_0, $z \approx N(0,1)$; reject H_0 at 5% if $z > 1.645$
- Results
  - before (Sep 2014 – 10 Jan 2024, $k = 85$): $\hat{\alpha} = 4.16$ ($s = 0.52$)
  - after (11 Jan 2024 – Sep 2026, $k = 24$): $\hat{\alpha} = 4.31$ ($s = 0.91$)
  - $z = 0.15$
- Placebo, S&P 500, 17 Sep 2014–18 Sep 2026; break 11 Jan 2024, $k = 58/16$ before/after: $2.82 \to 2.86$, $z = 0.03$

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A Non-Significant Result with Low Power: What We Cannot Conclude

- **Minimum detectable change**, one-sided 5% test with 80% power
  - $(1.645 + 0.842) \times 1.05 = 2.6$
  - 1.645: the one-sided 5% critical value; 0.842: the 80% quantile of $N(0, 1)$, for 80% power
  - $1.05 = \sqrt{0.52^2 + 0.91^2}$ is the SE of the difference
  - the Normal-distribution approximation is rough with 24 tail observations
  - the data cannot distinguish an unchanged tail from a large thinning of the tail
- A non-significant result does not show the absence of an effect
  - wait for more data or pool assets

MFM_ch1_ai_discovery

Project Seed

- **Question**: did the tail index of Bitcoin and Ethereum change with institutionalisation (futures 2017, spot ETFs 2024)?
  - raw and GARCH-standardised returns (shape versus scale)
  - loss and gain tails separately
  - Bitcoin uses the course series; Ethereum is not in the course data: document its source, the close time and time zone, and the currency (USD) before estimating
- **Replicate first**: the chapter's Hill table and the mini-case numbers
- **Pre-register**, committed before the results
  - H_0, H_1, the rule for k, break dates and bootstrap
- **Report**: the change-point test of Hoga (2017) as a Quantlet
  - estimates, both SEs, the minimum detectable change
- **Log**: prompts in `AI_USE.md`
  - every caught error (an i.i.d. SE, a date chosen from the data) in `AI_ERRORS.md`
  - reusable prompts (▸ Appendix) and checks (▸ Appendix)

What We Keep: the Size of Returns Is Forecastable, the Direction Hardly

Takeaways

- Returns are uncorrelated but not independent: their size is predictable
- Heavy tails are the rule: tail indices near 3
  - the fourth moment very likely does not exist
  - kurtosis-based inference fails
- Volatility is persistent (long memory or breaks)
  - in equity markets it reacts asymmetrically to losses
- Every test needs a reference distribution that the data may violate

Questions for the room

- Why does a mixture with σ_t drawn independently each day have heavy tails but no volatility clustering?
- Why does the S&P 500 statistic fall from $Q(10) = 91.6$ to $\tilde{Q}(10) = 19.0$ with robust standard errors?
- Why can the tail index be estimated with the Hill estimator when the kurtosis does not exist?

- Next chapter: are returns predictable at all? (Chapter 2)

Formula Sheet with Assumptions

Returns and annualisation

- $R_t = P_t/P_{t-\mathbf{1}} - 1$, $r_t = \ln(1+R_t)$, $r_t(k) = \sum_{j=\mathbf{0}}^{k-\mathbf{1}} r_{t-j}$, $R_{p,t} = \sum_i w_i R_{i,t}$
- $\text{CAGR} = (P_T/P_{\mathbf{0}})^{\mathbf{1}/Y} - 1$
- $\sigma_{\mathbf{ann}} = \sqrt{A}\,\hat{\sigma}$: uncorrelated returns, constant variance

Tests

- $JB = T\left[\hat{S}^{\mathbf{2}}/6 + (\hat{K}-3)^{\mathbf{2}}/24\right]$: $\chi^{\mathbf{2}}_{\mathbf{2}}$ under i.i.d. Normal
- $Q(m) = T(T+2)\sum_{k=\mathbf{1}}^{m} \hat{\rho}_k^{\mathbf{2}}/(T-k)$: $\chi^{\mathbf{2}}_m$ under i.i.d.
- $\tilde{Q}(m) = \sum_k \hat{\rho}_k^{\mathbf{2}}/\hat{\tau}_k$: $\mathbb{E}r^{\mathbf{4}} < \infty$, zero cross terms
- Hill $\text{SE}(\hat{\alpha}) \approx \hat{\alpha}/\sqrt{k}$: independent Pareto tail

Volatility

- Parkinson $\frac{\mathbf{1}}{\mathbf{4}\ln \mathbf{2}}\overline{(h-l)^{\mathbf{2}}}$, the bar a mean over days: driftless, no overnight jumps
- Mixture $K = 3\,\mathbb{E}[\sigma^{\mathbf{4}}]/\mathbb{E}[\sigma^{\mathbf{2}}]^{\mathbf{2}}$: σ_t independent of ε_t

Performance

- $SR = (\mu - r_f)/\sigma$ on simple returns: $\mu = A\bar{R}$, not the log mean $A\bar{r}$
  - $\ln(1+\text{CAGR}) \approx \mu - \sigma^{\mathbf{2}}/2$ (log-scale volatility drag)
  - $\text{SE} \approx 1/\sqrt{Y}$ for i.i.d. Normal returns
- $SoR = (\mu - r_f)/\sigma^{-}$
- $DD_t = P_t/\max_{s\le t} P_s - 1$; Calmar $= \text{CAGR}/|MDD|$

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Prices Before Computers, the Stock Ticker



Stock ticker designed by Thomas Edison

(Smithsonian collection)

Photo: Jaclyn Nash, Smithsonian National Museum of American History; public domain; Wikimedia Commons

- Before computers, trades reached brokers as prices printed on a paper **ticker tape**
- A daily close is one snapshot of this stream of prices
  - returns are built from such snapshots
- The choice of snapshot (close, fixing, a midnight snapshot) matters
  - as Chapter 0 showed
- Prices at the level of individual trades: market microstructure, Chapter 10

Back

The People Behind the Stylised Facts (1/2)



Carl Friedrich Gauss (1777–1855)

Painting: C.A. Jensen (1840); public domain; Wikimedia Commons

- The **Normal distribution**: the benchmark that returns violate



Louis Bachelier (1870–1946)

Photo: unknown author; public domain; Wikimedia Commons

- Prices as a random walk, in his thesis Bachelier (1900)



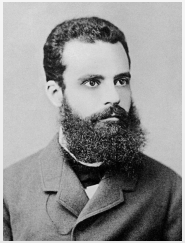
William S. Gosset, "Student" (1876–1937)

Photo: unknown author; public domain; Wikimedia Commons

- The Student-t distribution Student (1908), today a model for heavy tails

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The People Behind the Stylised Facts (2/2)



Vilfredo Pareto (1848–1923)

Photo: unknown author (1870s); public domain; Wikimedia Commons



Benoît Mandelbrot (1924–2010)

Photo: Rama (2007); CC BY-SA 2.0 France; Wikimedia Commons

- Pareto: **power laws** for large incomes, the ancestor of heavy-tail models
- Mandelbrot (1963): cotton prices
  - heavy tails, which the Normal distribution does not describe
- The exponent α of these power laws
  - is the tail index estimated with the Hill method

Back

Cont's Facts in Later Studies

- Ratliff-Crain, Van Oort, Koehler & Tivnan (2025)
  - re-test all 11 facts on intraday returns of the Dow 30 stocks, October 2018 – March 2019
  - conclusive evidence for 8 facts, no support for the remaining 3
- Chakraborti, Toke, Patriarca & Abergel (2011)
  - survey the empirical facts of returns and of the order book, from intraday to lower frequencies

Back

Annual Variance under Autocorrelation (1/2)

- The variance of a sum of h daily returns depends on the autocorrelations

$$\mathrm{Var}\Big(\sum_{j=1}^{h} r_{t+j}\Big) = h\sigma^2\Big[1 + 2\sum_{j=1}^{h-1}(1 - j/h)\rho_j\Big]$$

  - h: the horizon in days; σ^2: the daily variance; ρ_j: the autocorrelation at lag j
  - assumption: weakly stationary returns (constant mean and variance, autocovariances depending only on the lag)
  - without autocorrelation ($\rho_j = 0$) it is $h\sigma^2$, i.e. the $\sqrt{h}$ rule for volatility
- **Variance ratio** $VR(h) = \mathrm{Var}(h\text{-day sum})/(h\sigma^2)$
  - $VR(h) = 1$: no autocorrelation; $VR(h) > 1$: positive autocorrelation; $VR(h) < 1$: negative autocorrelation
  - volatility over horizon h is the $\sqrt{h}$-rule value multiplied by $\sqrt{VR(h)}$

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Annual Variance under Autocorrelation (2/2)

- Autoregression of order 1, AR(1), with $\rho_j = \rho^j$: $\sqrt{VR(h)} \to \sqrt{(1+\rho)/(1-\rho)}$ as h grows
  - ρ: the lag-1 autocorrelation, $|\rho| < 1$
  - EUR/RON, $\rho_1 = 0.17$: factor 1.19, annual volatility 4.4% $\to$ 5.3%
  - S&P 500, $\rho_1 = -0.079$: factor 0.92
- Using the sample lag-1 autocorrelation as an AR(1) parameter is an assumption
  - check the whole ACF (Seminar 1, A7)
  - variance-ratio tests: Chapter 2

The Null Hypothesis of JB, and Tests for Time Series

- JB is a Lagrange Multiplier test Jarque & Bera (1980)
  - under i.i.d. returns from the Normal distribution, $\sqrt{T}\,(\hat{S}, \hat{K} - 3) \to N\big(0, \mathrm{diag}(6, 24)\big)$, hence χ^2_2
  - $\mathrm{diag}(6, 24)$: a diagonal covariance matrix; $\hat{S}$ and $\hat{K}$ are asymptotically independent, with variances $6/T$ and $24/T$
  - JB is the sum of the two squared standardised statistics, hence χ^2 with 2 degrees of freedom
- GARCH-type dependence changes $\mathrm{Var}(\hat{S})$ and $\mathrm{Var}(\hat{K})$
  - a rejection mixes the marginal distribution and independence
- Bai & Ng (2005): tests of skewness, kurtosis and normality of the marginal distribution of a time series
  - studentised with HAC variances
  - their separate skewness and kurtosis tests (null $S = 0$ or $K = 3$ without normality) need $\mathbb{E}r^6 < \infty$ and $\mathbb{E}r^8 < \infty$

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The Robust Portmanteau Test and Its Assumptions

- $\tilde{Q}(m) = \sum_{k=1}^{m} \hat{\rho}_k^2 / \hat{\tau}_k \sim \chi_m^2$ Lobato, Nankervis & Savin (2001) needs $\mathbb{E}r^4 < \infty$
  - $\sim$: "is distributed as", here asymptotically
  - and zero cross terms: $\mathbb{E}[x_t^2 x_{t-k} x_{t-l}] = 0$ for two different lags $k \neq l$
  - then the $\hat{\rho}_k$ at different lags are asymptotically uncorrelated and the sum is χ_m^2
- Dalla, Giraitis & Phillips (2022) keep only the significant cross terms in the covariance of the $\hat{\rho}_k$
  - none for the S&P 500, BET-TR and Bitcoin
- $\hat{\tau}_k$ is the variance under $\rho_k = 0$
  - to compare a non-zero ρ_1 across periods use the HAC SE of the slope of r_t on r_{t-1} (Seminar 1, B11)
- ACF of $|r_t|$ and of r_t^2: Normal limits need $\alpha > 4$ and $\alpha > 8$ Davis & Mikosch (1998)

Back

Hill Estimator, Asymptotic Distribution and Alternatives

- Hill = maximum likelihood estimator of a Pareto tail above $u = X_{(k+1)}$, conditional on u (Seminar 1, A11)
- If $k \to \infty$ and $k/n \to 0$, and the tail is Pareto up to a second-order term

$$\sqrt{k}\,(\hat{\alpha}_k - \alpha) \;\to\; N(0, \alpha^2)$$

  - second-order term: a correction to the power law that vanishes as $x \to \infty$
  - consequence: $\mathrm{SE}(\hat{\alpha}_k) \approx \alpha/\sqrt{k}$, the formula in the main part
  - if k grows too fast, a bias term appears: the tail is only approximately Pareto
- Dependent data (GARCH-type): $\hat{\alpha}$ stays consistent
  - but extremes cluster and the variance exceeds α^2/k Hsing (1991)
- Choosing k: plot $\hat{\alpha}$ against $\log k$ ("alt-Hill" plot) Drees, Resnick & de Haan (2000)
  - the log scale stretches the region of small k, where the actual tail lies
- Regression alternative Gabaix & Ibragimov (2011)
  - log rank on log size, with rank shifted by $-\frac{1}{2}$: $\ln(i - \frac{1}{2}) = a - b \ln X_{(i)}$, $i = 1, \ldots, k$
  - the slope $\hat{b}$ estimates α; the shift by $\frac{1}{2}$ reduces the small-sample bias
  - $\mathrm{SE} = \hat{\alpha}\sqrt{2/k}$

Back

Two Tails and the Tail of $|r_t|$

- $P(|X| > x) = P(X > x) + P(X < -x)$
  - with indices α_- (losses) and α_+ (gains), $|X|$ has index $\min(\alpha_-, \alpha_+)$
- At the same k the estimates are not comparable
  - the k largest $|r_t|$ come from both tails
  - common threshold u: the lower of the two one-sided thresholds at $k = 2.5\%$ of n

	u (%)	k for $\lvert r_t \rvert$	$\hat{\alpha}$, $\lvert r_t \rvert$	$\hat{\alpha}_-$, losses	$\hat{\alpha}_+$, gains
S&P 500	2.17	508	2.90	2.85	2.96
BET-TR	1.79	153	2.54	2.20	3.06
Bitcoin	7.11	233	3.02	2.71	3.36
EUR/RON	0.54	295	2.28	2.63	2.04
Gold	1.98	513	3.11	2.97	3.28

- At a common u, the $|r_t|$ estimate lies between the two tails
  - Bitcoin's 4.2 at $k = 2.5\%$ reflects second-order bias
  - report both one-sided indices
  - Seminar 1, B6 uses $k = 2\%$ of n per tail: Bitcoin 2.89 / 3.62, BET-TR 2.19 / 3.08

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Which Tail Index Is the Chapter's Estimate?

- The chapter uses the fixed reference choice: $|r_t|$ at $k = 2.5\%$ of n (table in the main part)
- The common-threshold k for $|r_t|$ lies outside the 1–5% range of the main part
  - S&P 500 $508/9{,}245 = 5.5\%$, gold $513/9{,}559 = 5.4\%$, EUR/RON $295/5{,}352 = 5.5\%$, Bitcoin 5.3%, BET-TR 5.1%
  - a larger k lets less extreme days in, which can bias $\hat{\alpha}$ (choice of k)
- EUR/RON falls from 2.82 to 2.28
  - more than twice as many tail days (295 versus 133)
  - and the gain tail, $\hat{\alpha}_+ = 2.04$, is the heavier one: $|r_t|$ inherits the heavier tail
  - read the gap as threshold sensitivity, not as a second estimate of the same quantity
- The common-threshold table answers a different question: which tail, losses or gains, is heavier

Estimating d, the Formulas (1/2)

- Fourier frequencies $\lambda_j = 2\pi j/T$, $j = 1, \ldots, m$; the m lowest are used, here $m = T^{0.65}$
- The **periodogram**: the sample estimator of the spectral density

$$I(\lambda_j) = \frac{1}{2\pi T}\left|\sum_{t=1}^{T} x_t\, e^{-\mathrm{i}\lambda_j t}\right|^2$$

  - x_t: the demeaned series (here $|r_t|$ minus its mean); i: the imaginary unit
  - $I(\lambda_j)$: how much of the variation in x_t comes from the cycle of frequency λ_j
- Log-periodogram regression Geweke & Porter-Hudak (1983): regress $\ln I(\lambda_j)$ on $-2\ln(2\sin(\lambda_j/2))$
  - the reason: $\ln f(\lambda) \approx \ln G - 2d \ln \lambda$, and $2\sin(\lambda/2) \approx \lambda$ at low frequencies
  - the slope is $\hat{d}$, SE $= \pi/\sqrt{24m}$

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Estimating d, the Formulas (2/2)

- Local Whittle Robinson (1995): $\hat{d}$ minimises an objective built from the approximation $f(\lambda) \approx G\lambda^{-2d}$

$$\hat{d} = \arg\min_{d} \left[\ln\left(\frac{1}{m} \sum_{j=1}^{m} \lambda_j^{2d} I(\lambda_j) \right) - \frac{2d}{m} \sum_{j=1}^{m} \ln \lambda_j \right]$$

  - $\arg\min_d$: the value of d that minimises the expression; the constant G has been eliminated analytically
  - $\sqrt{m}(\hat{d} - d) \to N(0, \frac{1}{4})$, so $\text{SE} = 1/(2\sqrt{m})$
- Local Whittle is robust to non-Gaussianity
  - a larger m lowers the SE
  - but lets short-run (GARCH) dynamics bias $\hat{d}$ (Seminar 1, B12)

Markets in Detail

- **Crypto-assets**: an asset class with its own statistical profile Härdle, Harvey & Reule (2020)
  - tail, memory and moment factors classify cryptos as a separate asset class Pele et al. (2023)
  - the separation comes mainly through the tail factor
  - crypto modelling and the CRIX (CRypto IndeX): Härdle, Chen & Overbeck (2017); Chapter 16
- **EUR/RON**: BNR reference rate from 1 July 2005
  - the new leu (RON, 1 RON = 10,000 old lei)
  - long calm periods interrupted by sudden adjustments
  - EUR/RON market (interbank) quote series showed spurious negative autocorrelation (quote errors)
- **BET-TR**: the bank-asset tax was announced on 18 December 2018
  - adopted on 28 December 2018 as Government Emergency Ordinance (GEO) 114/2018
  - policy shocks produce single-day crashes without comparable rallies
- **Non-synchronous trading**: index constituents do not all trade at every moment
  - so the index reflects news with a delay
  - creating positive autocorrelation Campbell, Lo & MacKinlay (1997), ch. 3

Back

The Maximum Drawdown of a Brownian Motion

- Driftless Brownian motion of the log price with annual volatility σ over T years
  - here T is the horizon in years, not the number of days
- The maximum **log** drawdown $D = -\ln(1 + MDD) \geq 0$, the largest fall in the log price, has

$$\mathbb{E}[D] = \sqrt{\pi/2}\,\sigma\sqrt{T} \qquad \text{Magdon-Ismail, Atiya, Pratap \& Abu-Mostafa (2004)}$$

- $MDD = e^{-D} - 1$; $\sqrt{\pi/2} \approx 1.25$
  - without drift the expected maximum drawdown grows in proportion to $\sqrt{T}$
  - with a positive drift $\mathbb{E}[D]$ grows only like $\ln T$
- Gold 1990–2026: $\sigma = 16.3\%$, $T = 36.7$
  - benchmark $\mathbb{E}[D] = 1.24$ versus observed 0.592 (-44.7%) (Seminar 1, A2(b–e))
- Drawdowns and Calmar ratios from samples of different length are not comparable

Back

Sharpe Ratios beyond the Normal Distribution

- i.i.d., any distribution with $\mathbb{E}r^4 < \infty$ Opdyke (2007)
  - $\text{Var}(\widehat{SR}_d) \approx (1 - S\,SR_d + \frac{K-1}{4} SR_d^2)/T$
  - S, K: the skewness and kurtosis of daily returns; for the Normal distribution ($S = 0$, $K = 3$) this gives Lo's formula
  - negative skewness and high kurtosis raise the variance of $\widehat{SR}_d$ when $SR_d > 0$
  - here the SEs change by at most 0.02
- The Deflated Sharpe Ratio corrects for non-normality and multiple testing (Chapter 13)

	CAGR	Sharpe	Sortino	Calmar
S&P 500	8.7%	0.55	0.78	0.15
BET-TR	21.2%	1.34	1.87	0.68
Bitcoin	53.9%	0.98	1.45	0.65
EUR/RON	1.8%	0.43	0.65	0.12
Gold	6.7%	0.48	0.69	0.15

MFM_ch1_risk_measures

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Case Study, Estimation Details

- Targets (Sections 3.4–3.5 of the paper)
  - the 30-day implied volatility index of each index and the 9-day VIX
  - realised volatility of day $t+1$
  - volatility in decimals (0.03 = 3 volatility points)
- Kernels, eq. (3.9) in the paper: $K(\tau) = Z_{\alpha,\delta}^{-\mathbf{1}}(\tau+\delta)^{-\alpha}$
  - normalised so that $\sum_i K(i\Delta t)\Delta t = 1$ on the 1,000-day window
  - i: the age of the day, in days; $\Delta t = 1/252$: one day, in years
- Estimation: $(\beta_{\mathbf{0}}, \beta_{\mathbf{1}}, \beta_{\mathbf{2}}, \alpha_{\mathbf{1}}, \delta_{\mathbf{1}}, \alpha_{\mathbf{2}}, \delta_{\mathbf{2}})$ by nonlinear least squares on the training set
  - initialised using a lasso regression on features built with exponential kernels
- Our replication, started from the values in Table 3 of the paper (in parentheses)
  - $\hat{\beta}_{\mathbf{0}} = 0.052$ (0.057), $\hat{\beta}_{\mathbf{1}} = -0.090$ (-0.095)
  - $\hat{\beta}_{\mathbf{2}} = 0.868$ (0.82), $\hat{\alpha}_{\mathbf{1}} = 1.13$ (1.06)
- The 9-day VIX exists only from 2011 (shorter training set)
- The restricted models ($\beta_{\mathbf{1}} = 0$, $\beta_{\mathbf{2}} = 0$) are Seminar 1, C3

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Case Study, R^2 for All Indices (1/2)

Index	Implied volatility (paper, Table 2) R^2 train	R^2 test	Realised volatility (paper, Table 4) R^2 train	R^2 test
S&P 500 (VIX)	0.946	0.855	0.738	0.654
S&P 500 (9-day VIX)	0.876	0.914	–	–
Euro Stoxx 50	0.929	0.913	0.672	0.682
FTSE 100	0.925	0.870	0.650	0.617
DAX	0.934	0.918	0.722	0.557
Nikkei 225	0.890	0.800	0.563	0.504

- VIX parameters: $\beta_0 = 0.057$, $\beta_1 = -0.095$, $\beta_2 = 0.82$
- Across all targets $\alpha_2 \in [1.2, 2.3]$ (Tables 3 and 5 of the paper)
  - long memory near criticality (Section 3.2 of the paper)

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Case Study, R^2 for All Indices (2/2)

- Dropping Σ_t hurts the most
- The model beats two ARCH benchmarks and four published path-dependent models
  - Figures 3.8–3.9 and Appendix C of the paper
- Beyond this course (optional): its Markovian 4-factor version, driven by Brownian motion (Section 4.6 and Table 9 of the paper)
  - Markovian: the next value depends only on the current state (here, 4 factors), not on the whole path
  - gives estimated values of about 0.12–0.17 of the Hurst exponent H for daily realised volatility, as in data
  - H: the roughness exponent of a path; Brownian motion has $H = 0.5$, and $H < 0.5$ means a path "rougher" than Brownian motion
  - although it has no rough volatility ($H < 0.5$ built into the model), the Markovian version reproduces the small values of H estimated from daily data

The Discovery Loop, Reusable Prompts

Literature

- Find peer-reviewed work since 2015 on time variation or structural breaks in tail indices of asset returns, with tests valid under GARCH-type dependence.
- For each: DOI, estimator, test, data.
- Flag entries you are unsure of.

Hypothesis

- State competing hypotheses for a change in Bitcoin's tail after the 2024 spot ETFs.
- What does each imply for the Hill estimate of the tail index of raw and of GARCH-standardised returns?

Data and code

- Use daily Bitcoin closing prices in USD; log returns on its own calendar;
- Hill estimate of the tail index for $|r|$, with k = 2.5% of n, before and after 11 January 2024; moving-block bootstrap standard errors.

Robustness

- Write a pre-registered plan: k from 1% to 5% of n, loss and gain tails, GARCH-standardised residuals, block lengths 10 to 50 days, a placebo date and a placebo asset.

Critique

- Referee this design.
- With 24 tail observations after the break, what is the power against a change of 1 in the tail index?

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Mandatory Checks for the Researcher

- **Wrong return or price series**: reproduce the chapter's Hill table before anything else
- **Choice of k after seeing the result**
  - fix the rule ($k = 2.5\%$ of n) in advance
  - show the Hill plot for 1–5%
- **i.i.d. standard errors**: $\hat{\alpha}/\sqrt{k}$ ignores volatility clustering
  - compare with a block bootstrap
  - for S&P 500, 17 Sep 2014–10 Jan 2024: bootstrap SE 0.817 versus i.i.d. 0.370, hence 2.2 times larger
- **Break date chosen from the data**
  - use a date fixed by the event
  - or a sup-type change-point test with its own critical values Hoga (2017)
- **Bad ticks create fake tails**
  - list the ten largest $|r_t|$ and match each to a dated event
- **Low power**: report the minimum detectable change
  - 2.8 SEs of the difference for a two-sided 5% test with 80% power
  - 2.5 SEs for a one-sided test

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Practical Checklist for Empirical Work

- **Know your data**
  - price vs total-return index
  - adjusted vs raw prices
  - fixing vs traded price
  - official reference rates where possible (EUR/RON)
- **Clean before you model**
  - non-trading days, holidays, stale quotes, obvious errors
  - match the largest moves to real events
- **Use log returns** for single-asset dynamics, simple returns for portfolios
- **Report robust statistics**: quantiles, the number of tail days and the tail index $\hat{\alpha}$ next to kurtosis
- **Annualise carefully**
  - 252 vs 365 periods
  - the $\sqrt{A}$ rule needs uncorrelated returns with constant variance
- **Reproducibility**: code and data on GitHub, as Quantlets

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Further Study on Quantinar

Advanced courses on quantinar.com

- Statistics of Financial Markets
- Tukey's g- and h- transformations
- Applied Time Series Analysis with Python

Code, data and recommended reading

- All charts and tables: github.com/danpele/MFM/tree/main/Quantlets/Ch_01
- Recommended reading: Cont (2001); Franke, Härdle & Hafner (2019); Campbell, Lo & MacKinlay (1997), ch. 1–2

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Acronyms used in this material (1/2)

- **ACF**: AutoCorrelation Function
- **AI**: Artificial Intelligence
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARCH-LM**: ARCH Lagrange Multiplier (test)
- **BET-TR**: BET Total Return (index)
- **BNR**: Banca Națională a României — *the National Bank of Romania*
- **BVB**: Bursa de Valori București — *the Bucharest Stock Exchange*
- **CAGR**: Compound Annual Growth Rate
- **CI**: Confidence Interval
- **CLT**: Central Limit Theorem
- **CRIX**: CRypto IndeX
- **DAX**: Deutscher Aktienindex — *the German stock index*
- **EGARCH**: Exponential GARCH
- **ETF**: Exchange-Traded Fund
- **FTSE**: Financial Times Stock Exchange (FTSE Russell, index provider)
- **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- **GEO**: Government Emergency Ordinance
- **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HAR-RV**: Heterogeneous AutoRegressive model of Realised Volatility
- **JB**: Jarque–Bera (test)
- **LB**: Ljung–Box (test)
- **LM**: Lagrange Multiplier
- **MDD**: Maximum DrawDown
- **OHLC**: Open, High, Low, Close
- **PDV**: Path-Dependent Volatility
- **QQ**: Quantile–Quantile (plot)
- **RMSE**: Root Mean Squared Error
- **SE**: Standard Error
- **TSPL**: Time-Shifted Power Law (kernel)
- **US**: United States
- **VIX**: Cboe Volatility IndeX
- **VIX9D**: Cboe 9-Day Volatility Index

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